



Research article

Integrated fuzzy trust management with federated machine learning approaches for enhancing intrusion detection in IoT networks

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ABSTRACT

As the Internet of Things (IoT) expands, our environment is increasingly composed of vast networks of interconnected devices that generate large volumes of data. These smart devices improve efficiency and enhance the quality of human life; however, their diversity and scale create significant challenges in terms of trustworthiness and security. Traditional security solutions often struggle to handle the dynamic and complex threat landscape of IoT systems. To address these challenges, we propose a novel framework that integrates fuzzy trust management (FTM), the synthetic minority oversampling technique (SMOTE), and a federated support vector machine model (FTS-FedSVM). Fuzzy trust management enables accurate evaluation of trust by considering multiple factors, such as device behavior, network activity, and reputation, providing the adaptability needed for evolving threats. FedSVM allows distributed devices to collaboratively train models without sharing raw data, which reduces computational overhead while protecting privacy. This cooperative strategy ensures that even resource-constrained IoT devices benefit from the collective knowledge of the network. Extensive experimental evaluations demonstrate the effectiveness of various federated learning models. In contrast, the proposed Fed-SVM framework significantly outperforms these models, achieving an impressive 99.03% accuracy in binary classification (benign vs. malicious nodes) and 91.30% accuracy in multi-class attack detection, highlighting its robustness and superior learning capability compared to Federated Random Forest and Federated Decision Tree. It effectively mitigates threats such as botnets, ransomware, and network sniffing in heterogeneous IoT environments. Moreover, the framework reduces computational power consumption by approximately 80%, demonstrating its suitability for sustainable IoT security.

1. Introduction

The Internet of Things (IoT) is a fast-growing paradigm that links a wide range of devices, using integrated sensors and networks to enhance daily life and improve living standards [1]. It facilitates the connection of sensor equipped devices, enabling seamless data

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collection and transmission. Moreover, its ability to interact with internal and external conditions aids decision-making processes. IoT integrates smart devices into homes and cities, creating a comprehensive information system. However, to realize its full potential, addressing security concerns is imperative [2].

Ensuring the security of IoT systems is an ongoing challenge, and researchers and practitioners are continuously devising solutions that can adapt to ever-changing threats while safeguarding the availability, integrity, and confidentiality of IoT data and services [3,4]. The different machine learning models use to mitigate the different types of malicious attacks [5,6] protect data privacy [7–9] build trust management, support federated learning, and involve blockchain technology [10,11] to improve IoT security. Additionally, it improves IoT device authentication, makes adaptive security possible, guarantees respect for legal and regulatory standards, and promotes industry collaboration. The strategy intends to achieve a complete and flexible security solution that tackles the specific difficulties in protecting IoT networks [12–14], promotes confidence among network users, and secures the long-term availability and integrity of IoT data and services. Additionally, it encourages conformity with legal and regulatory standards [15].

We propose a novel hybrid model based on federated support vector machine, SMOTE and fuzzy trust management (FTS-FedSVM) to secure IoT networks, improving trust among network users. The proposed approach calculates the trustworthiness of each node using direct, indirect, and mutual trust techniques. The parameters used to evaluate trust include credibility, recommendation, and robustness. Based on these parameters, a trust degree is assigned to each node, which contributes to building a secure architecture for IoT devices. A single trust rating is generated using either a consensus method or a weighted average of the trust values. The Federated SVM model then classifies IoT nodes as either malicious or benign to detect potential security threats. This intelligent system helps automatically identify and counter multi attacks in the IoT network more easily.

The proposed security mechanism for IoT networks has integrated SVM Federated Learning with Trust Management. It is meant to enhance security by accommodating the varying risks, trust establishment between people and devices as well as applying aggregated trusted approach. The security approach better helps in adapt quickly to the evolving threat landscape and it protects from running afoul of legal and regulatory requirements. It facilitates industry [16] cooperation with standardized security practices and standards to address IoT end-to-end security. At a higher level, the objective is to overcome security issues in IoT networks and improve data accuracy for all of those billions of connected devices. This would definitely improve the understanding of any system as shown in Fig. 1, six layers IoT network architecture proposed below. IoT Devices, Data Processing, Analytics User Engagement Security, Business Process which makes a fully modular solution for clear and concise requirements.

1. **Device Layer:** At the core of the architecture, the Device Layer showcases a range of IoT devices and sensors. Each type of device is clearly labeled to make identification straightforward.
2. **Communication and Middleware Layer:** This layer connects devices by utilizing arrows and labels to denote communication protocols. It combines communication and middleware capabilities. Edge computing hardware and gateways are examples of middleware components.
3. **Application Layer:** The Application Layer keeps its original organization and is where user interface components (UI) like dashboards, mobile apps, and web interfaces are hosted. These interfaces engage in communication with the middleware and communication layers.
4. **Data Processing and Analytics Layer:** By combining data processing and analytics activities into a single layer, it is clear how vital a role they play in processing and evaluating data gathered from Internet of Things (IoT) devices. Icons for analytics engines, machine learning components, and data processing tools represent the processing capabilities.
5. **IoT Platform and Security Layer:** This layer merges the IoT Platform Layer with the Security and Privacy Layer, handling the core management functions for devices and data. Icons such as locks and shields illustrate the integrated security measures and features.
6. **Cloud/Server and Business Layer:** This unified layer, blending the Cloud/Server Layer with the Business Layer, encompasses cloud servers and data centers. It illustrates data storage, processing power, and the integration of business processes.

The simplified 6-layer IoT architecture streamlines system design, making it easier to understand and customize for various projects. However, implementing and managing an IoT network comes with challenges such as security, privacy, scalability, regulatory compliance, and human factors. To navigate these challenges effectively, a comprehensive strategy is essential, incorporating robust security measures, meticulous planning, and stakeholder engagement.

1.1. Objective of FTS-FedSVM approach

This paper aims to propose a comprehensive security model for IoT networks, incorporating innovative approaches to enhance trust, threat detection, and legal compliance. Key objectives include addressing security challenges and promoting stakeholder participation.

1. Explore IoT's transformative potential while addressing critical security and trustworthiness issues caused by its rapid growth.
2. To develop a real-world IoT trust management dataset that captures various types of attacks on IoT devices.
3. Develop a system using fuzzy logic to manage trust uncertainty in IoT devices, SMOTE for dataset balancing enabling better decision-making for trust establishment and maintenance.

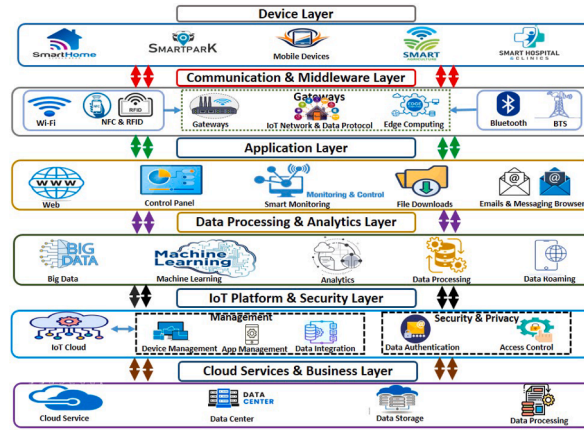


Fig. 1. Architecture of IoT network.

- Integrate fuzzy trust management with a hybrid federated SVM learning model to enhance IoT device security through collaborative learning and threat detection.
- Our FTS-FedSVM model requires less computational power than other approaches due to its lightweight trust mechanism, while also enabling more accurate decision-making.

1.2. Organization of research article

The rest of this article is structured as follows: [Section 2](#) introduces the core concepts of the proposed approach and offers a detailed review of existing research on IoT network security. [Section 3](#) describes the Hybrid Federated SVM, Random Forest, and Decision Tree models in detail, covering their design, training methodology, and application in intrusion detection. It also elaborates on trust management, the role of fuzzy logic in handling uncertainty, and the application of SMOTE for data balancing. [Section 4](#) introduces the proposed simulation framework for securing IoT environments, emphasizing the integration of trust management within the security architecture and demonstrating its effectiveness through a real-world use case. Provide the proposed approach comparison with the other previous approaches. Finally, [Section 5](#) summarizes the key contributions of the FTS-FedSVM approach, highlights promising research directions, and concludes.

2. Literature review

The interaction of several domains, leading to novel approaches and challenging problems, describes the present state of technology. The field of federated learning, a decentralized machine learning approach that shows potential in various applications, has attracted a lot of attention in recent years. This review of the literature highlights a few of these research, each of which sheds light on a different aspect of the developing disciplines.

In a 2022 study, a Federated Learning-based method was proposed for anomaly detection in IoT networks [17]. This approach performs federated training across GRU (Gated Recurrent Unit) models using decentralized, on-device data to identify network intrusions, while periodically sharing updated model weights with a central FL server. However, the method has limitations due to its reliance on a single central server, which can lead to communication delays, network congestion, inefficient resource utilization, and reduced effectiveness in detecting attacks not represented in the training data.

The article in 2022 introduced AutoTrust, a recurrent neural network-based trust management technique for the Internet of Cloud Things security(IoCT) and privacy issues, offering an autonomic solution for trust management [18]. A newly formed dataset containing 70,000 values is used to assess the suggested mechanism and compare it to others already in use. The limitations of a proposed mechanism, include the required resources for trust degree transmission, additional computing energy for multiple-layer integration, and assessment against potential IoT attacks.

In another recent study in 2022, the paper discussed the use of Industrial Control Systems (ICSs), AI, and Big Data in Smart Manufacturing, and the challenges in creating safe ICSs [19]. The study describes the FedeX architecture for Smart Manufacturing in Industrial Control Systems as a Federated Learning-Based Explainable Anomaly Detection architecture. It explains its automatic threshold determination technique utilizing Support Vector Data Description and assesses its performance on liquid storage and SWAT datasets. However, the limitations are to emphasize the privacy and security concerns in communication channels between remote zones and the cloud, suggesting safe encryption techniques to prevent tap-in and improve FedeX architecture.

In 2019, a study introduced Perceptron Detection (PD), a method combining perceptron and K-means algorithms to identify malicious nodes in IoT networks [20]. The research employed a multi-mix-attack model alongside PD to improve detection accuracy

Table 1
Structure of literature review.

Ref	Approach	Contribution	Limitation
[17]	Decentralized FL using GRUs	Federated Learning with Gated Recurrent Units (GRUs) for improved model training.	May not be suitable for IoT devices with limited resources.
[18]	RNN-Based Trust Management	Utilizes Recurrent Neural Networks for detecting malicious behaviors.	Primarily focuses on Internet of Cloud Things (IoCT), which might limit applicability to other IoT environments.
[19]	FedeX: Federated Learning-Based Explainable	Employs Support Vector Data Description (SVDD) for automatic threshold determination in Federated Learning.	Privacy and security concerns in remote communication channels.
[20]	Perceptron and K-means Algorithms	Proposes a detection mechanism using Perceptron and K-means algorithms.	May not be directly applicable to dynamic IoT networks.
[21]	Machine Learning and Network QoS	Evaluates IoT device trust using machine learning techniques and network Quality of Service (QoS) metrics.	Focuses on specific IoT devices and may not generalize to all IoT environments.
[22]	Decentralized Federated Learning	Combines LSTM and GRUs for decentralized Federated Learning.	Potential privacy breaches or attacks on central servers or local devices.
[23]	Decentralized Learning and Resource-Intensive Network Centralization	Addresses challenges in decentralized Federated Learning and integrates Reinforcement Federated Learning (R-FL).	Complexity in IoT infrastructure management.
[24]	Decentralized Federated Learning	Enhances Federated Learning approaches with a decentralized framework.	Limited to a single dataset (N-BaIoT) and does not address potential ethical implications.
[25]	Blockchain-Based Secure Data Collaboration	Integrates Federated Deep Learning with Blockchain for secure data collaboration.	Context-specific to IoT, may not generalize to other data collaboration scenarios.
[26]	Gradient-Based Vulnerability Detection	Proposes a gradient-based algorithm to detect malicious nodes by analyzing model update rankings.	No practical implementations and limited effectiveness against sophisticated attacks.
[27]	Federated DNN Trust management	FedTrust uses federated learning and a novel 19-parameter trust dataset to efficiently detect malicious IoT nodes with 87% accuracy.	Its performance vary across IoT environments, and reliance on dedicated servers can add communication overhead and potential points of failure.
[28]	Federated Trust Fuzzy logic	Privacy preserving by federated trust and fuzzy logic for single class mitigation.	Limited by small scale deployment, and narrow comparison with recent methods.
[29]	Federated Learning	Federated learning for UAV-enabled IoT networks, improving accuracy, reducing loss, and enhancing security with high scalability and anomaly detection.	Its performance under complex attacks and real-world conditions is untested, and scalability for large networks may need optimization.
[30]	Multifactor fuzzy trust FL	MFTE approach predicts malicious 80% in MEC.	Need to reduce computational power and improve security.
[31]	Zero-Trust FL	Trust-aware FL aggregation using Dirichlet distribution for IoT security with 83.91% accuracy.	Scalability constraints.
[32]	Hierarchical FL with Trust Management	Secured IoT metaverse transactions, achieving 93% accuracy and 25% computational reduction.	High complexity; unsuitable for large, dynamic networks.

by 20-30%. The study highlighted the importance of strong security measures and the susceptibility of IoT networks to insider threats. However, by not distinguishing between different attack types, the method's accuracy can be limited, particularly in scenarios with low path diversity.

In order to create a behavioral model and quantify trust, the study in 2021 presents an IoT device trust evaluation approach that uses machine learning and network QoS [21]. It explores major challenges in trust evaluation in IoT, including metrics and computational approaches, and handles security concerns while identifying trust mechanisms. The suggested approach, which relies on network QoS data, may not always be accurate, demand a lot of processing power, miss some new or undiscovered assaults, and not work with all IoT devices.

In 2021, the authors explored the integration of IoT for smart applications in healthcare, smart homes, and industrial systems [22]. They proposed a decentralized federated learning approach for anomaly detection in IoT networks using LSTM (Long Short-Term Memory) and GRU (Gated Recurrent Unit) models to improve detection accuracy and reduce false alarms. However, the study, which focused on a specific Modbus network dataset, highlighted that little work has been done on developing a decentralized, communication-efficient architecture for anomaly detection across IoT networks.

For privacy and security in crucial IoT infrastructure systems, the study from 2022 recommends a decentralized learning and resource-intensive network centralization-addressed R-FL system [23]. The study proposed an integrated R-FL system and federated learning for data centralization in healthcare and e-commerce, assessing system trust based on end-device ratings. This article's

limitation is that the proposed blockchain network management model would need a lot of processing power, might not be ideal for all IoT applications, and might need highly technical knowledge.

In 2022, the authors of the article described how to identify malware in IoT devices using federated learning, a decentralized machine learning technique [24]. The N-BaIoT dataset was used to assess the efficacy of the proposed framework, which uses federated learning to handle security concerns, manage sensitive data, and identify malware on IoT devices. The suggested architecture and federated learning technique may have several drawbacks including scalability, communication cost, the requirement for a sizable dataset, and resilience against malevolent actors.

The study in 2020 introduced a federated deep learning and blockchain-based secure data collaboration architecture for multiparty computing in the IoT context [25]. The architecture, which has three layers data, control, and application was demonstrated to be useful in a real IoT situation. Despite potential challenges caused by data requirements, processing resources, and secure data centers.

In a study published in 2021, a novel algorithm called MANDERA was introduced [26]. It can automatically discover the fact that some gradient nodes are Byzantine (i.e., they produces large/meaningless gradients) using their values as input. The recommended model was demonstrated in contrast to some reputable aggregating tactics with MANDERA. Mander develops a weighted evaluation matrix from gradient space to differentiate positive and negative gradients. While, ideas such as MANDERA (which converts the gradient values into a ranking space) have shown improvements in detection and model accuracy are not helpful since it lack practical utility against real-world sophisticated attacks.

The proposed FedTrust approach introduces a federated DDNlearning-based trust management framework that identifies malicious and compromised IoT nodes using a novel dataset with 19 trust parameters across knowledge, experience, and reputation. By organizing IoT devices into communities with dedicated servers, FedTrust reduces computational overhead and achieves an accuracy of 87% in detecting abnormal behaviors. However, its performance may vary across different IoT environments, and reliance on dedicated community servers could introduce additional communication overhead and potential points of failure [27].

This study introduces a privacy-conscious federated learning system, FLTM, which employs fuzzy logic to support an effective and explainable trust assessment in fog-enabled Wi-Fi mesh networks. Empirical evidence indicates that FLTM outperforms both centralized and existing federated learning paradigms, and naturally reduces single-class bias in the generation of fuzzy-based initial labels. However, the research is limited by the use of a small number of nodes within a single fog node, the manual creation of the fuzzy rule ensemble, and a limited evaluation of modern federated learning systems. Future studies are needed to scale FLTM to large-scale, heterogeneous tasks, use adaptive or auto-generated fuzzy inference systems to support dynamic single-class distributions and class imbalance, investigate more advanced multi-model learning approaches, and critically analysis practical implementation issues in fog and Internet of Things systems [28].

In this article [29] author introduces a federated learning framework for UAV-enabled IoT networks, improving accuracy, reducing loss, and enhancing robustness under attacks, with high scalability and anomaly detection rates that boost security. While promising, the framework's performance under complex attack scenarios and real-world conditions remains untested. Additionally, further optimization may be needed to address scalability challenges for extremely large networks.

This paper introduces MFTE, a scheme of multifactor fuzzy-based trust assessment of FL nodes in MEC, which involves behavior, reliability, history, data integrity, and reputation. T-S fuzzy filtering model helps to reduce dishonest advice and uncertainty of trust. Experiments demonstrate that MFTE is more robust and detects well in the presence of over 80% malicious nodes (and malicious nodes occur in 30 percent case) with global accuracy. However, scalability constrained by increasing computational power and communication overheads and encourages future edge-terminal collaboration and two-way trust assessment [30].

In research paper [31] the author proposed a zero-trust federated learning system that uses Dirichlet-based dynamic trust evaluation of client selection and aggregation to obtain an accuracy of 83.91 percent. The main value of the research is a betrayal-conscious min-max optimization model which has been re-formulated as a convex problem, and with this, it can be solved efficiently. The obtained results of the simulation indicate that the trust is more accurate and the resistance to failures and inside attacks is greater. However, the related computational cost and reliance on accurate updates on trust limit scalability, thus highlighting the importance of future studies on lightweight and scalable trust mechanisms.

This paper presents a new hybrid fuzzy logic-based trust management system, called HFL-TM, which incorporates decentralized federated learning to provide dynamic and runtime trust evaluation and effective fraud detection in the context of IoT-enabled Metaverse systems. The main value of the framework is the synergistic fusion of the fuzzy trust analysis and lightweight hierarchical federated learning aggregation which provides a significant robustness against a range of attack vectors and at the same time decreases computational energy usage up to 25 percent. However, the framework effectiveness depends on the exact definition of the fuzzy rules, and is also subject of extra coordination overhead when using in large scale highly dynamic network applications [32].

2.1. Research gap in IoT network security

Despite the literature review section highlighting numerous issues regarding internet security and privacy, as well as the use of different machine learning models and other techniques to address these challenges, the problem remains unsolved. Some major research findings were identified during the literature review. Most researchers rely on existing deep learning models such as RNNs, LSTMs, and GRUs, which often require high computational and communication resources, making it difficult to deploy in large-scale and resource-constrained IoT environments. The models [17,22,23] presented are not efficient in terms of privacy and security. Moreover, the tested work [18,24,29] relies on a single dataset or within a strictly defined scope, cannot be applied to heterogeneous IoT environments or real-world attack scenarios. Centralized or hierarchical forms of trust management are often adopted, introducing communication overhead, scalability issues, and single points of failure [27,31]. Lastly, adaptive and interpretable decision-making

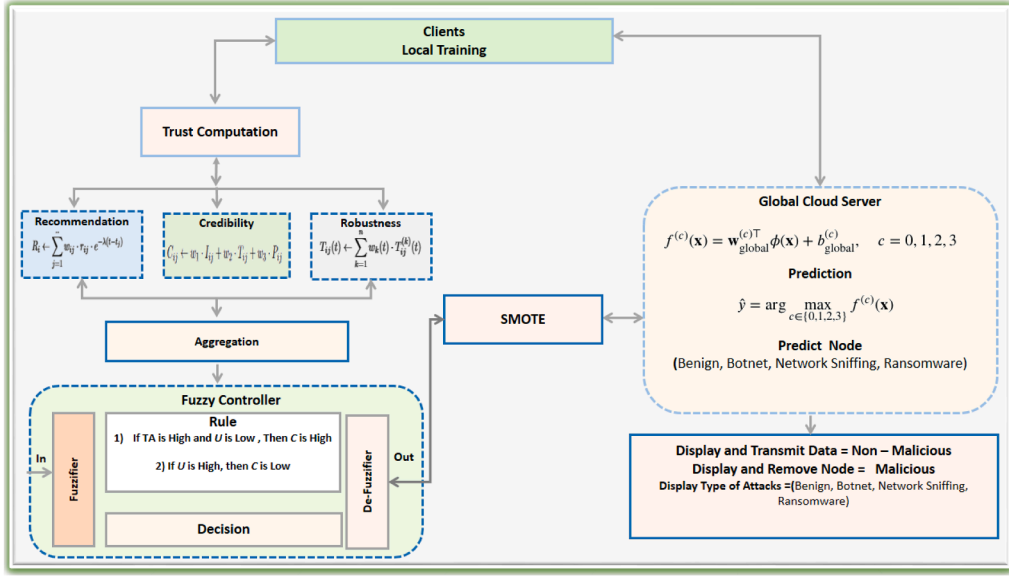


Fig. 2. Proposed methodology for IoT network.

remains minimal, during the fuzzy logic implementation, unbalancing nature of the data are generally not considered [19,26]. To address these limitations, this paper proposes a federated SVM-based trust model that integrates SMOTE and fuzzy logic, enabling the detection of malicious nodes in both binary and multi-class frameworks. The model is scalable, reliable, and less computationally intensive for heterogeneous IoT systems, while preserving privacy.

3. Methodology

The proposed methodology aims to strengthen the security of IoT devices through an effective trust management approach. It evaluates several parameters such as recommendation, credibility, and robustness to determine the trust level of each IoT device. The overall trust value is calculated by combining these parameters credibility, recommendation and robustness, while fuzzy logic helps reduce uncertainty during this process. To ensure balanced data, the SMOTE technique is applied before using a federated SVM model to classify IoT nodes as either malicious or non-malicious. This approach helps identify and prevent security threats before they can exploit system weaknesses, ensuring a more reliable and secure IoT environment. Overall, the proposed framework provides a robust and efficient solution to strengthen IoT systems against evolving cyber threats, as shown in Fig. 2.

3.1. Research questions

In our research, we address the challenges of security and privacy associated with IoT networks. Our proposed methodology effectively counters these issues. The research questionnaire is provided below.

1. How can IoT's transformative potential be harnessed while addressing critical security and trustworthiness challenges through real-world datasets capturing diverse attacks?
2. How can fuzzy logic and SMOTE be combined with federated SVM learning to manage trust uncertainty, improve decision-making, and enhance collaborative threat detection in IoT networks?
3. How does the FTS-FedSVM model compare with existing IoT security methods in terms of accuracy and computational efficiency?

3.2. Trust degree evaluation in IoT networks

The "Trust Degree Evaluation in IoT Networks" focuses on assessing the trustworthiness of devices within IoT environments. We utilize key parameters such as recommendation, robustness, and credibility, along with different trust models: direct, indirect, and mutual trust. By aggregating these trust factors, we aim to provide a comprehensive and reliable trust degree assessment for secure IoT operations, as elaborated below.

Node Interaction Model: The interaction between nodes i and j is represented as I_{ij} , which is a composite function based on key factors: communication reliability (R_{ij}), data integrity (D_{ij}), and mutual trust (T_{ij}).

Credibility Evaluation: The credibility of node j as perceived by node i (C_{ij}) is determined by combining several factors:

$$C_{ij} = w_1 \cdot I_{ij} + w_2 \cdot T_{ij} + w_3 \cdot P_{ij} \quad (1)$$

where in Eq. (1) P_{ij} represents the historical behavior or reputation of node j as observed by node i , and w_1 , w_2 , and w_3 are weights that adjust the impact of each factor on the credibility assessment.

Quantifying Mutual Trust: To measure mutual trust (T_{ij}) between nodes, a mathematical framework is used that considers historical interactions, shared goals, and established protocols. This framework tracks trust evolution over time by incorporating factors such as interaction frequency, consistency, and adherence to protocols. Techniques from game theory and machine learning may also be used to improve trust measurement, making it more robust against adversarial actions and evolving network conditions.

Recommendations of Trustworthy Nodes: To calculate the recommendation parameter R for nodes in an IoT network, we use a weighted averaging approach based on feedback from trusted neighboring nodes. The recommendation score for node i , denoted as R_i , is the weighted average of recommendations from its neighbors. Each neighbor's recommendation is weighted by factors such as proximity, trust level, or expertise (w_{ij}), with the actual recommendation value being r_{ij} . The formula for R_i is:

$$R_i = \frac{\sum_{j=1}^n w_{ij} r_{ij}}{\sum_{j=1}^n w_{ij}} \quad (2)$$

In Eq. (2), n is the number of neighboring nodes providing feedback.

Robustness Dynamic Trust Evaluation: The trust value T_{ij}^t representing the confidence of device i in device j at time t is iteratively updated through a Bayesian-inspired recursive formulation:

$$T_{ij}^t = \lambda T_{ij}^{t-1} + (1 - \lambda) \tilde{T}_{ij}^t \quad (3)$$

where in Eq. (3):

- λ denotes the temporal weighting coefficient that controls the influence of historical trust compared to recent observations.
- $\tilde{T}_{ij}^t$ represents the freshly computed trust estimate derived from the latest interaction outcomes.

Adaptive Reputation System: An adaptive reputation evaluation mechanism updates the reliability score of device i at time t as follows:

$$R_i^t = \alpha R_i^{t-1} + (1 - \alpha) \tilde{R}_i^t \quad (4)$$

where in Eq. (4):

- α denotes the memory retention factor, determining how much influence past reputation has compared to the newly computed score.
- $\tilde{R}_i^t$ represents the instantaneous reputation derived from the most recent feedback and node interactions.

Trust Fusion Mechanism: To derive an aggregated trust measure, multiple trust evaluations are integrated through a dynamic weighted fusion process:

$$T_{ij}^t = \sum_{k=1}^n \omega_k^t \tilde{T}_{ij}^{(k),t} \quad (5)$$

where in Eq. (5) :

- $\tilde{T}_{ij}^{(k),t}$ denotes the trust value inferred from the k^{th} evidence source at time t .
- ω_k^t represents the adaptive weight corresponding to each evidence source, reflecting its reliability and contribution to the overall trust evaluation.

Recommendation Analysis of Trust Weight: To enhance the model, we factor in the credibility of recommending nodes and the recency of recommendations. By considering each recommending node's credibility (C_j), we refine the recommendation score R_j .

Refinement with Credibility and Recency: We introduce a time decay factor λ to give more weight to recent recommendations and reduce the impact of older ones.

Incorporating Time Decay Factor: Normalization techniques, such as min-max normalization, are used to ensure recommendation scores are comparable across different nodes.

To integrate these aspects into a unified trust assessment framework, we propose:

$$T_{ij} = w_1 \cdot C_{ij} + w_2 \cdot T_{ij}^{dynamic} + w_3 \cdot R_{ij} + w_4 \cdot T_{ij}^{fusion} \quad (6)$$

where in Eq. (6):

- C_{ij} is the credibility assessment.
- $T_{ij}^{dynamic}$ is the dynamically evaluated trust.
- R_{ij} represents the recommendation parameter.
- T_{ij}^{fusion} is the trust fusion outcome.

The weights w_1 , w_2 , w_3 , and w_4 adjust the influence of each component based on the specific needs and characteristics of the IoT system described in [Algorithm 1](#). This framework allows customization to prioritize different aspects of trust according to system requirements and environmental factors.

Fuzzy Sets: For each linguistic variable, we define fuzzy sets with appropriate membership functions. For example: - Trust Aggregation (TA): Low, Medium, High.

$$TA = \mu_{low}(TA) + \mu_{medium}(TA) + \mu_{high}(TA) \quad (7)$$

Eq. (7) represents the aggregation of trust values using fuzzy membership functions, where each μ denotes the degree of membership of TA in the corresponding fuzzy set (low, medium, or high).

- Uncertainty (U): Low, Medium, High.

$$U = \mu_{low}(U) + \mu_{medium}(U) + \mu_{high}(U) \quad (8)$$

Eq. (8) captures the fuzzified uncertainty level, showing how uncertainty is distributed across low, medium, and high fuzzy sets to handle imprecision in trust evaluation.

Fuzzy Rules: Based on expert knowledge or data-driven analysis, fuzzy rules are established to map the relationship between *Trust Aggregation* (TA) and *Uncertainty* (U) toward a resulting *Certainty* (C) value.

Rule 1: If trust aggregation is high and uncertainty is low, then certainty is high.

$$C = \mu_{high}(TA) \wedge \mu_{low}(U) \quad (9)$$

Eq. (9) expresses the fuzzy inference rule using the logical AND ($\wedge$) operator, indicating that a high trust value combined with low uncertainty leads to a high degree of certainty in the system's decision.

Rule 2: If uncertainty is high, then certainty is low.

$$C = \mu_{low}(U) \quad (10)$$

Eq. (10) captures the inverse relationship between uncertainty and certainty, implying that higher uncertainty directly reduces the system's confidence level in the trust evaluation outcome.

Fuzzy Inference System: Using the defined fuzzy sets and rules, we implement a fuzzy inference system to compute the degree of *Certainty* (C) based on *Trust Aggregation* (TA) and *Uncertainty* (U).

$$C = \max(\min(\mu_{high}(TA), \mu_{low}(U)), \min(\mu_{low}(U))) \quad (11)$$

Eq. (16) represents the fuzzy inference process using the *max-min composition* method. The min operator models the fuzzy AND relationship between input membership functions, combining the degrees of trust aggregation and uncertainty. The max operator aggregates the outcomes of all fuzzy rules to produce the final certainty level. Thus, the equation determines the highest degree of certainty derived from the interaction of all rule-based evaluations.

Membership Functions and Fuzzy Rules: These are determined based on expert knowledge or data analysis, and their specifics would depend on the context of the IoT system and the available data as shown in [Algorithm 2](#).

SMOTE-Augmented Dataset: After applying SMOTE to handle class imbalance, the dataset is augmented with synthetic samples. Let the feature matrix and label vector be updated in [Eq. \(12\)](#).

$$\mathbf{X}_{SMOTE} = \begin{bmatrix} \mathbf{X}_{original} \\ \mathbf{X}_{synthetic} \end{bmatrix}, \quad \mathbf{y}_{SMOTE} = \begin{bmatrix} \mathbf{y}_{original} \\ \mathbf{y}_{synthetic} \end{bmatrix} \quad (12)$$

Trust Aggregation: The overall trust for node i is calculated by combining information from both the original and synthetically generated samples:

$$TA_i = \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} T_{ij}, \quad \forall i \in \mathbf{X}_{SMOTE} \quad (13)$$

In [Eq. \(13\)](#), $\mathcal{N}_i$ denotes the set of neighbors of node i , incorporating contributions from synthetic minority samples produced using the SMOTE technique.

Fuzzy Inference Rules: Expert knowledge is used to define fuzzy rules that map *Trust Aggregation* (TA) and *Uncertainty* (U) to a corresponding *Certainty* (C) measure.

$$C_i^{(1)} = \mu_{high}(TA_i) \wedge \mu_{low}(U_i), \quad \forall i \in \mathbf{X}_{SMOTE} \quad (14)$$

Eq. (14) illustrates that a combination of high trust aggregation and low uncertainty yields a high certainty score, using the logical AND operator ($\wedge$).

$$C_i^{(2)} = \mu_{low}(U_i), \quad \forall i \in \mathbf{X}_{SMOTE} \quad (15)$$

Eq. (15) captures the scenario where elevated uncertainty leads to a low certainty value, reflecting diminished confidence in the trust assessment.

Fuzzy Inference System: The final degree of certainty is computed by aggregating all fuzzy rules using the max-min method:

$$C_i = \max \left(\min(\mu_{high}(TA_i), \mu_{low}(U_i)), \min(\mu_{low}(U_i)) \right), \quad \forall i \in \mathbf{X}_{SMOTE} \quad (16)$$

Algorithm 1: Computation of trust parameters in IoT nodes.

Input: Communication reliability R_{ij} , data integrity D_{ij} , mutual trust T_{ij} , historical performance P_{ij} , parameters α, β, λ , weights $\omega_1, \omega_2, \omega_3, \omega_4$, recommendation feedback r_{ij} , credibility of recommenders C_j , time t

Output: Credibility C_{ij} , Adaptive Trust T_{ij}^t , Recommendation R_{ij}

Step 1: Credibility Estimation

if C_{ij} is evaluated then

$$I_{ij} = \text{Evaluate trust using } R_{ij}, D_{ij}, \text{ and } T_{ij}$$

$$C_{ij} = \omega_1 I_{ij} + \omega_2 T_{ij} + \omega_3 P_{ij}$$

$$T_{ij}^t = \alpha T_{ij}^{t-1} + (1 - \alpha) \tilde{T}_{ij}^t$$

Step 2: Recommendation Evaluation

if R_i is evaluated then

$$R_i^t = \beta R_i^{t-1} + (1 - \beta) \tilde{R}_i^t$$

$$R_{ij} = \frac{\sum_{j=1}^n \omega_{ij} r_{ij}}{\sum_{j=1}^n \omega_{ij}}$$

$$R_i = \frac{\sum_{j=1}^n \omega_{ij} r_{ij} C_j}{\sum_{j=1}^n \omega_{ij} C_j}$$

$$R_i = \sum_{j=1}^n \omega_{ij} r_{ij} e^{-\lambda(t-t_j)}$$

Step 3: Robustness Computation

if T_{ij} is evaluated then

$$T_{ij}^t = \alpha T_{ij}^{t-1} + (1 - \alpha) \tilde{T}_{ij}^t$$

$$T_{ij}^t = \sum_{k=1}^n \omega_k^t \tilde{T}_{ij}^{(k),t}$$

return Credibility C_{ij} , Adaptive Trust T_{ij}^t , and Recommendation R_{ij} .

Eq. (16) applies the max-min composition method, where the min operator computes the degree of certainty within each rule, and the max operator aggregates all rules to produce the final certainty value. This ensures a balanced and robust evaluation across the SMOTE-augmented dataset.

3.3. Mathematical model of federated random forest learning

Data Pre-Processing: Let $\mathbf{X} \in \mathbb{R}^{m \times n}$ denote the feature matrix of the dataset, where each row $\mathbf{x}_i \in \mathbb{R}^n$ corresponds to a data instance and each column corresponds to a feature. The $\mathbf{y} \in \{0, 1\}^m$ represent the label vector indicating the class of each data instance, where 1 denotes a malicious node and 0 denotes a non-malicious node.

Model Training: At each local node i , a decision tree is trained using its local dataset,

$$\mathcal{D}_i = \{(\mathbf{x}_{i1}, y_{i1}), (\mathbf{x}_{i2}, y_{i2}), \dots, (\mathbf{x}_{in_i}, y_{in_i})\};$$

$$\text{DT}_i = \text{DecisionTree}(\mathcal{D}_i) \quad (17)$$

Decision Tree Splitting Criteria: The decision tree algorithm employs impurity measures such as Gini impurity or entropy to determine optimal splits at each node.

Model Aggregation: The locally trained decision trees are aggregated to construct the federated random forest model $\text{RF}(\mathcal{D})$:

$$\text{RF}(\mathcal{D}) = \frac{1}{N} \sum_{i=1}^N \text{DT}_i \quad (18)$$

Weighted Aggregation: Optionally, weights can be assigned to each local decision tree based on performance metrics or other relevant criteria.

Prediction: Given a new data instance $\mathbf{x}$, the federated random forest model predicts its corresponding label $\hat{y}$:

$$\hat{y} = \text{RF}(\mathcal{D})(\mathbf{x}) \quad (19)$$

Algorithm 2: Fuzzy inference system.

Input : Trust Aggregation (TA), Uncertainty (U)
Output: Degree of Certainty (C)

Define linguistic variables: TA , U ;
Define fuzzy sets for TA and U with appropriate membership functions;
Define fuzzy rules based on expert knowledge or data analysis:

1. If TA is **High** and U is **Low**, then C is **High**.
2. If U is **High**, then C is **Low**.

while convergence criteria not met **do**
 Update membership functions and rules based on new data;
 Perform fuzzy inference to compute updated C ;
 Check convergence criteria;
end
return Degree of Certainty (C);

Table 2
Symbols and their descriptions.

Symbol	Description
I_{ij}	Interaction between nodes
R_{ij}	Communication reliability
D_{ij}	Data integrity
T_{ij}	Mutual trust
C_{ij}	Credibility of j by i
P_{ij}	Historical behavior
w_1, w_2, w_3, w_4	Weights for factors
R_i	Recommendation score
r_{ij}	Recommendation value
$T_{ij}(t)$	Trust level at time t
α	Forgetting factor
$\hat{T}_{ij}(t)$	New trust estimate
$R_i(t)$	Reputation score
β	Reputation decay factor
$\hat{R}_i(t)$	Updated reputation score
$T_{ij}^{(k)}(t)$	Trust based on evidence k
TA	Trust aggregation value
U	Uncertainty in trust
C	Degree of certainty
X	Feature matrix
y	Label vector
D_i	Local dataset at node i
DT_i	Decision tree at node i
RF	Federated random forest
$\hat{y}$	Predicted label
N	Number of nodes
T	Number of trees
λ	Time decay factor

Model Evaluation: The performance of the federated random forest model can be assessed using standard evaluation metrics, including accuracy, precision, recall, F1-score, and the area under the receiver operating characteristic curve (AUC).

Hyperparameter Tuning: Key hyperparameters, such as the number of trees (T), maximum depth of each tree, minimum number of samples per leaf, and the size of the feature subset, can be optimized through approaches like grid search or randomized search. A detailed description of the Federated Random Forest algorithm is provided in [Algorithm 3](#).

3.4. Federated SVM for IoT attack detection

In this approach, we use **Federated Support Vector Machine (SVM)** to detect various types of attacks in IoT networks. The algorithm functions within a federated learning setup, in which each IoT device (client) trains a local SVM model using its own dataset. Only the model updates are transmitted to a central server, which combines these updates to refine the global model. The resulting global model is then employed for attack classification.

Algorithm 3: Federated random forest for prediction of malicious nodes.

Input: IoT datasets $\mathcal{D}_i = \{(\mathbf{x}_{ij}, y_{ij})\}$ for clients $i \in \{1, \dots, N\}$, number of trees T
Output: Final global Random Forest model RF_{global} for malicious node detection

Step 1: Data Preprocessing

Split dataset $\mathbf{X}$ and labels $\mathbf{y}$ into local datasets $\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_N$ for each client.

Step 2: Local Model Training

foreach client $i \in \{1, 2, \dots, N\}$ **do**

 Train local decision trees $\text{DT}_i^{(1)}, \dots, \text{DT}_i^{(T)}$ using dataset $\mathcal{D}_i$.

Step 3: Model Aggregation

Aggregate all local trees to construct the global Random Forest:

$$\text{RF}_{global} \leftarrow \frac{1}{N} \sum_{i=1}^N \text{RF}_i$$

Step 4: Prediction

for each new data instance $\mathbf{x}$ **do**

 Predict label $\hat{y}$ using RF_{global} :

$$\hat{y} = \text{RF}_{global}(\mathbf{x})$$

Step 5: Output

return Final global random forest model RF_{global} and predicted labels $\hat{\mathbf{y}}$.

The global model M_{global}^t is updated at each communication round t by averaging the local models M_i^t from all clients. The update rule is given in Eq. (20) as elaborated below.

$$M_{global}^t = \frac{1}{N} \sum_{i=1}^N M_i^t \quad (20)$$

where:

- N is the number of clients (IoT devices).
- M_i^t is the local model at client i after round t .
- M_{global}^t is the aggregated global model at round t .

This model averaging technique ensures that the global model M_{global}^t reflects the collective knowledge of all clients, while preserving data privacy. Each client only shares model parameters hyperplanes, not the raw data.

Attack Prediction: After completing the federated training, the final global model M_{global}^T is used to classify IoT nodes into one of the following categories:

{Benign, Ransomware, Network Sniffing, Botnet}

The classification is done based on the decision function $f(\mathbf{x})$, which computes a score for each possible class. The class with the highest score is selected as the predicted attack type for each IoT node. The decision function for SVM is defined as:

$$f(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b \quad (21)$$

where in Eq. (21):

- $\mathbf{w}$ is the weight vector (defines the hyperplane).
- $\mathbf{x}$ is the feature vector representing the IoT node.
- b is the bias term.

The final attack prediction is made by selecting the class with the highest decision function score:

$$y_{pred} = \arg \max_{c \in \{\text{Benign, Ransomware, Network Sniffing, Botnet}\}} f(\mathbf{x}) \quad (22)$$

The different symbol used in Eq. (22) are shown below.

- y_{pred} is the predicted label for an IoT node.
- $f(\mathbf{x})$ is the decision function that provides the scores for each class.

Algorithm 4: Federated SVM for multi-class IoT attack detection.**Input:** IoT datasets $\mathcal{D}_i = \{(\mathbf{x}_{ij}, y_{ij})\}$ for clients $i \in \{1, \dots, N\}$, number of communication rounds T **Output:** Final global model M_{global} for attack classification**Step 1: Initialization**Initialize global SVM model M_{global}^0 .**Step 2: Federated Training****for** $t = 1$ **to** T **do** **foreach** client $i \in \{1, 2, \dots, N\}$ **do** Receive M_{global}^{t-1} from server. Train local SVM M_i^t on dataset $\mathcal{D}_i$. Send updated model parameters ΔM_i^t to server.

Server aggregates updates:

$$M_{global}^t \leftarrow \frac{1}{N} \sum_{i=1}^N M_i^t$$

Step 3: Attack PredictionUse final global model M_{global}^T to classify each IoT node as:

{Benign, Ransomware, Network Sniffing, Botnet}.

return Final global model M_{global}^T .

The mathematical modeling illustrates how the Federated SVM operates to classify IoT nodes as either benign or malicious. In cases identified as malicious, the model further determines the specific type of attack targeting the IoT network. Algorithm 4 provides a detailed description of the operational steps and decision-making process involved in this prediction framework.

3.5. Decision tree for IoT attack detection

In the experiment, the Decision Tree algorithm is employed to classify IoT nodes into different attack categories. A decision tree is a hierarchical structure in which each internal node represents a test on an attribute (such as packet size or trust score), each branch corresponds to the result of the test, and each leaf node assigns a class label, indicating whether the node is benign or corresponds to a specific type of attack.

Equation for Decision Tree: At each node, the decision tree partitions the data based on a feature x_j to either maximize information gain or minimize impurity (e.g., Gini impurity or entropy). The decision rule at each node can be formulated as shown in Eq. (23).

$$f(\mathbf{x}) = \begin{cases} c_1, & \text{if } \mathbf{x} \in \text{Leaf Node 1} \\ c_2, & \text{if } \mathbf{x} \in \text{Leaf Node 2} \\ \vdots & \\ c_n, & \text{if } \mathbf{x} \in \text{Leaf Node n} \end{cases} \quad (23)$$

where:

- $c_1, c_2, \dots, c_n$ are the classes (e.g., benign, ransomware, network sniffing, botnet).
- $\mathbf{x}$ is the feature vector of the IoT node.

The decision tree predicts outcomes by starting at the root node and following branches according to the feature values, ultimately reaching a leaf node, which determines the assigned class label.

Attack Prediction: After the decision tree is trained, the attack prediction is made by using the decision function $f(\mathbf{x})$, which routes the input feature vector $\mathbf{x}$ down the tree until it reaches a leaf node. The class label at the leaf node is the predicted attack type as shown in Algorithm 5.

3.6. Mechanism for federated learning to identify non-IID trust management data

To address the confines posed by artificially partitioned datasets in federated learning, we formulate a trust-aware system to be installed in the real-world Internet of Things (IoT) setting. The proposed mechanism is able to handle the issue of non-identically distributed (non-IID) and non-independent data distribution, communication constraints, convergence behavior, and client heterogeneity.

The mechanism consists of four main steps:

Algorithm 5: Decision tree for multi-class IoT attack detection.**Input:** IoT datasets $\mathcal{D}_i = \{(\mathbf{x}_{ij}, y_{ij})\}$ for clients $i \in \{1, \dots, N\}$ **Output:** Trained Decision Tree model T for attack classification**Step 1: Initialization**Initialize an empty decision tree T .**Step 2: Training Decision Tree****foreach** client $i \in \{1, 2, \dots, N\}$ **do** Train a decision tree T_i on dataset $\mathcal{D}_i$. Store the trained tree T_i .**Step 3: Combine Trees**Aggregate individual models from each client into a final decision tree T_{final} . This can be done through model averaging or majority voting based on tree decisions.**Step 4: Attack Prediction**Use the final decision tree T_{final} to classify each IoT node as:

{Benign, Ransomware, Network Sniffing, Botnet}.

return Final decision tree model T_{final} .

1. **Trust Score Evaluation:** The trust score is determined as a range $T_k \in [0, 1]$ and is attributed to each client k depending on historical contributions, the quality of the data, the regularity of the model updates, and the trustworthiness of communication. This process guarantees that the clients with untrustworthy or malevolent conduct have reduced impact over the global model.

$$T_k = f(\text{consistency, accuracy, behavior, communication reliability}) \quad (24)$$

2. **Non-IID Quantification:** Each client's data distribution P_k is compared with the global or cluster-level distribution P using KL-divergence to capture realistic data skew.

$$\text{Non-IID}_k = D_{\text{KL}}(P_k \parallel P) \quad (25)$$

This step of the procedure identifies clients with significantly skewed data distribution, which allows the use of adaptive weighting schemes that reduce bias in further model updates.

3. **Trust-Weighted Filtering and Aggregation:** Model updates from clients are aggregated using a weight that combines trust and non-IID skew:

$$\alpha_k = \frac{T_k}{1 + \text{Non-IID}_k} \bigg/ \sum_{j=1}^K \frac{T_j}{1 + \text{Non-IID}_j} \quad (26)$$

$$w_{\text{global}} = \sum_{k=1}^K \alpha_k w_k \quad (27)$$

Clients with higher levels of trust and representative datasets are weighted with an augmented influence and clients with low levels of trust or strong data skew are weighted with a reduced weight.

4. **Communication and Heterogeneity Awareness:** Each client's participation is adjusted based on bandwidth B_k and computation time t_k , ensuring efficient communication and reducing round latency:

$$w_{\text{global}} = \sum_{k=1}^K \frac{\alpha_k / t_k}{\sum_{j=1}^K \alpha_j / t_j} w_k$$

This ensures that faster, trusted clients contribute more, while slow or unreliable clients have minimal impact on the training process, illustrated in [Algorithm 6](#).

3.6.1. Proposed mechanism for performance evaluation

This mechanism addresses critical real-world IoT challenges:

- **Non-IID Data:** By measuring data skew and combining it with trust scores, the approach minimizes bias between dissimilar client data.
- **Communication Overhead:** Client participation is adjusted according to available bandwidth, reducing unnecessary communication and improving the effectiveness of iterative processes.
- **Convergence Behavior:** Even with heterogeneous clients, trust-weighted and skew-aware aggregation stabilizes updates and achieves faster convergence.
- **Client Heterogeneity:** Weighted aggregation accounts for differences in computation time and reliability, ensuring fair and efficient model updates.

Algorithm 6: Trust-based federated learning for non-IID IoT data.

Input : K clients, R rounds, initial model w^0 , learning rate η , trust sensitivity β , datasets $\{D_k\}$, computation times $\{t_k\}$, bandwidths $\{B_k\}$

Output: Trained global model w^R

for $r = 1$ **to** R **do**

for $k = 1$ **to** K **in parallel do**

// Compute trust and Non-IID

$T_k = f(\text{Con}, \text{AC}, \text{BE}, \text{C.R})$

$\text{NonIID}_k = D_{\text{KL}}(P_k \parallel P)$

// Local model update

$w_k^r = w^{r-1} - \eta \nabla F_k(w^{r-1})$

// Compute weighted aggregation with heterogeneity

$\alpha_k = \frac{T_k / (1 + \text{NonIID}_k)}{\sum_{j=1}^K T_j / (1 + \text{NonIID}_j)}$

$\alpha_k = \frac{\alpha_k / t_k}{\sum_{j=1}^K \alpha_j / t_j}$

$w^r = \sum_{k=1}^K \alpha_k w_k^r$

// Update trust scores

$T_k^{r+1} = T_k^r \cdot \left(1 - \beta \|w_k^r - w^r\|\right)$

return w^R ;

Overall, this trust-based system enables effective federated learning in realistic IoT settings, where clients with reliable, representative data have the greatest influence on the global model, and the negative impact of bias or unreliable contributions is also reduced.

3.7. Ablation study on the different component of proposed model

To explain this issue, we have included a conceptual ablation study [Table 3](#) which outlines the effects of removing each of the components. The research proves that trust management is an important component to ensure reliable client participation and robust aggregation in cases where client data are non-IID and adversarial manipulated; otherwise, federated learning is vulnerable to malicious updates and exhibits unpredictable convergence behavior. SMOTE addresses the extreme class imbalance that exists in the datasets of IoT intrusion, and its absence creates biased decision boundaries and worsening minority class detection performance. The implementation of a Federated SVM is further supported by a direct comparison to other federated classification paradigms, and thus it can be seen that simpler models do not provide the necessary robustness and generalization ability. Notably, the ablation test shows that the effects of these elements are operational, as opposed to incremental. The elimination of any one of the elements will result in a failure at a specific point of the pipeline, be it the representation of the data, its aggregation reliability, or its classification, resulting in the poor or distorted model performance. Thus, the improvements in performance which have been observed are the result of a synergistic combination of trust management, SMOTE, and Federated SVM, and not a result of a single parameter change.

4. Experimental setup and results discussion

The trust-based security framework for IoT was evaluated through simulation using Jupyter Notebook. Its performance in managing tasks and mitigating potential attacks was systematically assessed. The trust management approach measures the presence of malicious nodes, while machine learning algorithms are applied to detect them. The simulations were conducted on a Core i9 system with a 1 TB SSD and 16 GB of RAM. Analysis of the simulation results was performed using the Keras and TensorFlow libraries within Jupyter Notebook. Further details regarding the simulation configuration and parameters are provided in [Table 2](#).

Table 3
Conceptual ablation analysis of the proposed FTS-FedSVM framework.

Configuration	Trust	SMOTE	FedSVM	Expected Behavior	Limitation
Baseline FL	✗	✗	✗	Performs standard federated learning but struggles with class imbalance.	Vulnerable to attacks, sensitive to non-IID data, and prone to biased predictions
FedSVM	✗	✗	✓	Enables privacy-preserving SVM training among clients	Performance affected by imbalanced data malicious clients
Trust + FedSVM	✓	✗	✓	Robust aggregation mitigates attacks, improves trustworthiness	Minority classes remain under represented,transient in trust, limiting fairness
Proposed FTS-FedSVM	✓	✓	✓	Combines trust, oversampling,Fuzzy and federated SVM for robust, fair, and privacy-preserving learning	Effectively addresses class imbalance and security; minimal limitations

4.1. Dataset description for proposed FTS-FedSVM approach

The dataset consists of 30,000 rows and contains attributes that are essential for analyzing trust management and intrusion detection in networked systems. Each record includes a time stamp that logs the exact occurrence of an event, a source node and its associated IP address representing the origin of communication, and a target node that identifies the destination of the interaction. To measure the reliability of network entities, the dataset incorporates trust values in terms of recommendation, credibility, and robustness. Furthermore, it categorizes nodes as either benign or malicious, where malicious nodes are associated with different types of cyberattacks such as network sniffing, ransomware, and botnet infections. This combination of temporal, structural, trust-related, and attack-specific information makes the dataset a rich resource for building and testing advanced machine learning models, particularly federated learning-based intrusion detection systems and trust management frameworks. Its comprehensive nature enables researchers to study normal versus abnormal network behavior, evaluate anomaly detection techniques, and design resilient security mechanisms for distributed environments. The dataset is publicly available on Kaggle at the provided link <https://www.kaggle.com/datasets/kagglecomsohrab/ntrusion-detection-in-iot-networks>.

4.2. Fuzzy stabilization in the dataset

The uncertainty in trust management metrics is an essential challenge to address in dynamic systems. Trust is further enhanced by the use of fuzzy logic that allows a more flexible and continuous expression of trust as opposed to the discrete binary or multiclass representations inherent in traditional schemes. Traditional approaches often possess a sudden swings in the measure of trust, thus creating instability and affecting the reliability and robustness of the assessment. Conversely, fuzzy logic defines trust as a scale of membership grades (e.g., low to high), thus, making it easy to incorporate uncertainty in the decision-making process. Instead, fuzzy logic establishes trust as a range of membership grades (e.g., low to high) and, thus, enables the introduction of the element of uncertainty into the evaluation process. One of the major benefits of fuzzy logic is that it is able to reduce the sharp fluctuations in the value of trust as time progresses, thus avoiding sharp rises and falls in the value of trust hence stable and consistent updates in the value of trust. The results of the fuzzy-logic approach are more stable and resilient trust assessments than more traditional approaches, as demonstrated in Fig. 3. This resilience is especially important in changing environments like the Internet of Things and distributed systems, in which trust needs to keep up with the changing interactions and conditions.

4.3. SMOTE-based data balancing for optimized model training

Prior to applying the Synthetic Minority Over-sampling Technique (SMOTE), the dataset exhibited an imbalance between the Benign and Malicious classes. Specifically, there were 1389 samples of the benign class (label 0) and only 611 samples of the malicious class (label 1). This class imbalance could negatively affect the performance of machine learning models, as they may become biased toward predicting the majority class.

To mitigate this issue, SMOTE was applied to the dataset. The technique generates synthetic samples for the minority class (malicious) by interpolating between existing instances. After applying SMOTE, the dataset achieved a balanced class distribution, with 1389 samples in both the Benign and Malicious classes. This balanced dataset ensures that the model treats both classes equitably, reducing bias toward the majority class and improving the accuracy and fairness of predictions, as illustrated in Fig. 4.

4.4. Federated learning model for prediction

In this study, three federated machine learning models Federated Support Vector Machine (SVM), Federated Random Forest, and Federated Decision Tree are employed to differentiate between benign and malicious activities in IoT networks. These models are designed to address two key challenges: the class imbalance in the dataset and the distributed nature of IoT systems, where maintaining data privacy and security is essential. By using federated learning, the framework keeps raw data on local devices, reducing privacy

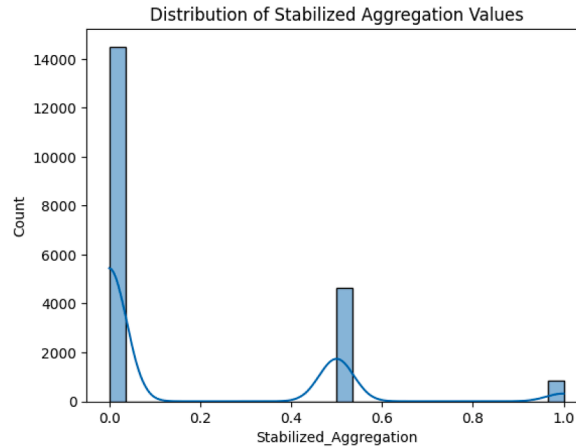


Fig. 3. Fuzzy stabilization of data.

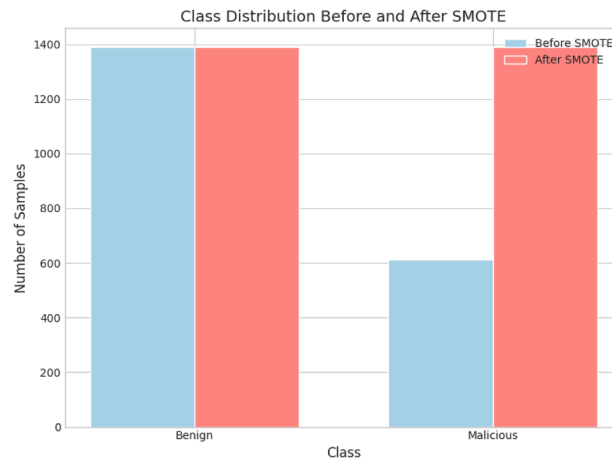


Fig. 4. SMOTE dataset balancing for prediction.

risks and complying with data protection requirements, while enabling collaborative training of models across heterogeneous IoT nodes. Our methodology follows a two phase classification strategy. In the first phase, a binary classifier is employed to differentiate between malicious and benign traffic. In the second phase, a multi-class classifier is applied to further categorize malicious traffic into specific attack types such as ransomware, botnets, and network sniffing. This hierarchical classification design enhances detection accuracy by reducing false positives and providing granular insights into the threat landscape. Furthermore, the federated approach improves model scalability and robustness, ensuring adaptability to dynamic, large-scale IoT deployments. By minimizing communication overhead and mitigating single points of failure, our models achieve high reliability in real-world scenarios. Collectively, these federated models not only improve classification accuracy but also offer an efficient and privacy-preserving solution to address the rapidly evolving cybersecurity threats in IoT ecosystems.

4.5. Federated random forest for binary threat detection in IoT

Accuracies of Clients and Global Model: The federated training of the Random Forest model was carried out across five distributed IoT clients, each contributing 280 local samples for the learning process. This setup reflects a balanced yet heterogeneous training environment, where data characteristics vary across clients due to non-identically distributed (non-IID) patterns inherent in IoT networks. The local training phase yielded accuracies of 82.67% (Client 1), 82.33% (Client 2), 76.67% (Client 3), 83.00% (Client 4), and 83.33% (Client 5). These results indicate that the model maintains high discriminative power for classifying benign versus malicious nodes, though the variability across clients underscores the challenges posed by skewed data distributions. The relatively lower performance of Client 3 (76.67%) suggests potential class imbalance or increased noise in the local dataset, which can degrade model precision at the edge level. Nonetheless, the ability of all clients to achieve above 75% accuracy illustrates the baseline robustness of the Random Forest architecture, even when trained locally with limited data. Following the federated aggregation phase using the Federated Averaging (FedAvg) algorithm, the global accuracy increased to 84.50%. This improvement reflects the synergistic effect of federated learning: by aggregating knowledge from diverse and distributed sources, the global model captures a richer

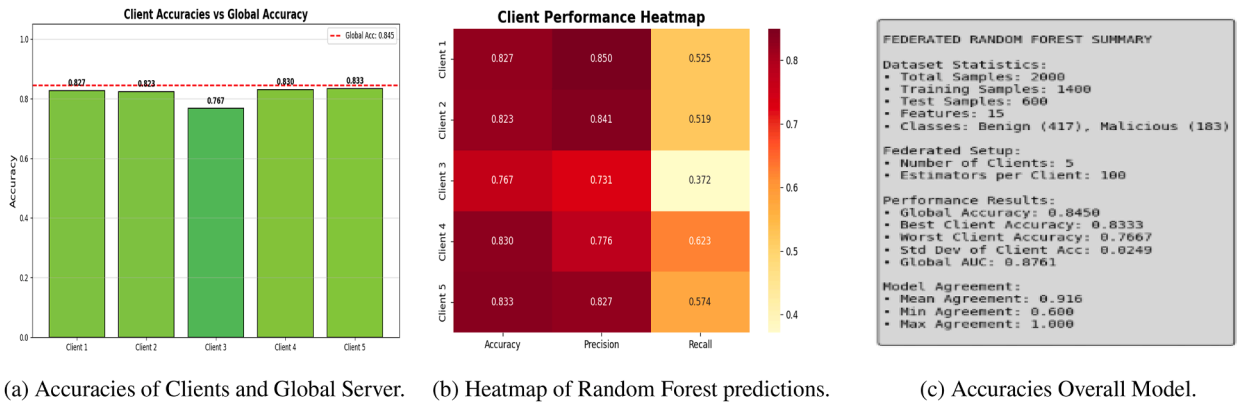


Fig. 5. Parameters comparison of the federated random forest models.

representation of the underlying data distribution. The global performance not only surpasses the mean local accuracy of 81.60% but also reduces the variance in predictive capability observed across individual clients, as shown in Fig. 5(a). The heatmap visualization illustrated in Fig. 5(b) displays the performance of the model, highlighting the accuracy achieved by each client as well as the overall precision and recall metrics in Fig. 5(c) presents these results, providing a clear overview of how effectively the model distinguishes between different classes.

ROC Curve Analysis of Federated Random Forest Binary Classifier: The ROC curve analysis, shown in Fig. 6(a), illustrates the performance comparison between five individual client models and the global Federated Random Forest model for classifying benign and malicious IoT nodes. Each client trained a local Random Forest model using 280 samples, resulting in AUC values of 0.840 (Client 1), 0.864 (Client 2), 0.810 (Client 3), 0.836 (Client 4), and 0.851 (Client 5). While all clients demonstrated strong discriminative capability with AUC values above 0.80, variability was observed, particularly in Client 3, likely due to data imbalance or noise in its dataset. The global model, aggregated using Federated Averaging, achieved the highest AUC of 0.876, with its curve lying above most client curves, confirming the advantage of federated learning in improving generalization. This demonstrates that the Federated Random Forest effectively integrates distributed knowledge across heterogeneous clients, outperforming local models and providing robust classification capability.

To further evaluate the model, the Precision-Recall (PR) curve of the federated model, depicted in Fig. 6(b), highlights its performance under class-imbalanced conditions. The curve demonstrates that the model maintains high precision across a wide range of recall values, effectively reducing false positives while accurately detecting malicious nodes. The Average Precision (AP) score of 0.805 reinforces the model's ability to balance precision and recall. Unlike ROC curves, which can sometimes give overly optimistic estimates on imbalanced datasets, the PR curve offers a more accurate evaluation by emphasizing the minority (malicious) class.

Comparing the two evaluation metrics, the global model's ROC AUC of 0.876 reflects its overall discriminative ability across thresholds, while the PR AP of 0.805 emphasizes its precision-recall trade-off in detecting the critical minority class. Together, the results from Fig. 6(a) and (b) demonstrate that the Federated Random Forest not only achieves superior generalization compared to local models but also maintains high reliability in precision-recall trade-offs, making it an effective solution for detecting malicious activity in distributed IoT environments.

Fig. 6(c) illustrates the prediction confidence distribution of the Federated Random Forest model for intrusion detection in IoT networks. The green bars represent correctly classified nodes, while the red bars denote misclassified ones. The figure shows that correct classifications are spread across lower to moderate confidence levels, whereas incorrect predictions cluster around the 0.4-0.5 confidence range, indicating uncertainty in distinguishing malicious traffic. This suggests that while the model is effective overall, refining decision boundaries could further reduce misclassifications and strengthen IoT security.

Confusion Matrix Analysis of Federated Random Forest Binary Classifier: The confusion matrix illustrates the performance of the Federated Random Forest model in classifying IoT nodes as either benign or malicious. It shows that the model correctly identified 407 benign nodes (True Negatives), indicating its effectiveness in recognizing normal network activity while minimizing false alarms. Furthermore, it successfully detected 100 malicious nodes (True Positives), which is the primary objective for ensuring network security. However, the model was not flawless, producing 10 False Positives where benign nodes were incorrectly flagged as malicious, which could lead to inefficient resource usage. Most critically, there were 83 False Negatives, representing actual malicious nodes that were misclassified as benign; these missed threats pose a significant security risk. To provide a more detailed assessment of the model's performance, the classification report in Table 4 presents the precision, recall, and F1-scores for each class. For the benign class, the model achieved a precision of 0.83 and a recall of 0.98, yielding an F1-score of 0.90 across 417 samples. In the case of the malicious class, the precision was 0.91, indicating that predictions of malicious activity were generally accurate, while the recall was 0.55, showing that a number of malicious nodes were not detected, consistent with the False Negative count in the confusion matrix. The resulting F1-score for the malicious class was 0.68 over 183 samples. Overall, the model reached an accuracy of 0.84 over 600 test instances. The macro-average F1-score was 0.79, reflecting balanced performance across both classes, and the weighted-average F1-score was 0.83, influenced by the larger number of benign samples. These results indicate that the Federated

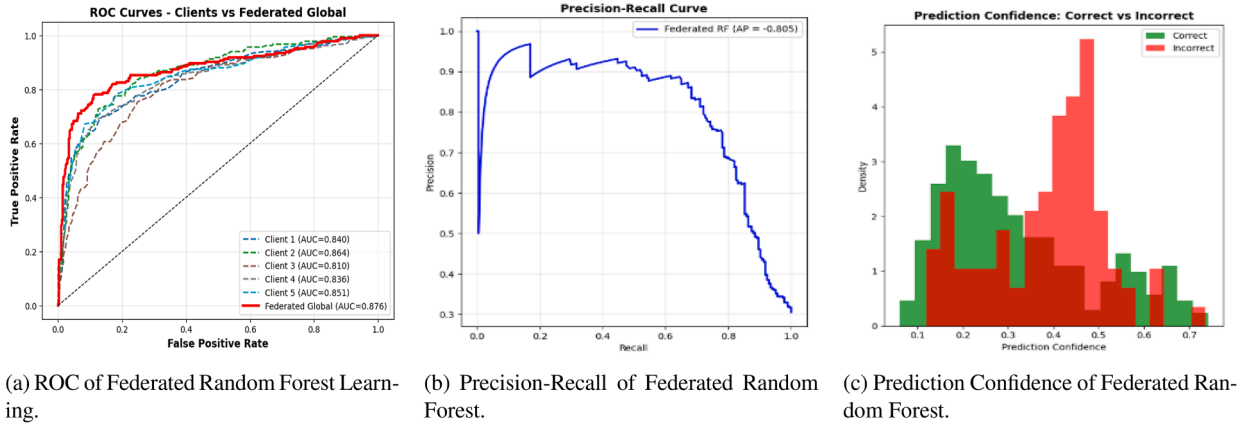


Fig. 6. Performance of the federated random forest binary classifier.

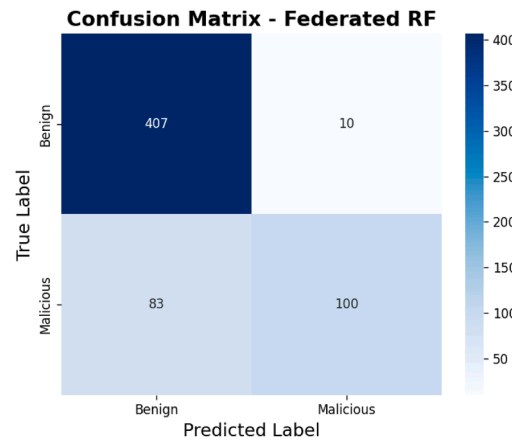


Fig. 7. Confusion matrix for federated random forest (Binary Classification).

Table 4

Detailed classification report of the model.

Class	Precision	Recall	F1-Score	Support
Benign	0.83	0.98	0.90	417
Malicious	0.91	0.55	0.68	183
Accuracy		0.84		600
Macro Avg	0.87	0.76	0.79	600
Weighted Avg	0.85	0.84	0.83	600

Random Forest model performs well in overall accuracy and reduces false alarms, though improvements are needed to enhance recall for malicious node detection. As shown in Fig. 7 and detailed in Table 4, the model demonstrates strong potential for IoT intrusion detection while highlighting areas for further optimization in detecting malicious nodes.

4.6. Federated SVM learning for binary threat detection in IoT

Accuracies of Clients and Global Model: Fig. 8(a) presents the comparison of client model accuracies against the global accuracy of the Federated SVM classifier for IoT intrusion detection. Each client model consistently achieved an accuracy of around 0.989, while the aggregated global model outperformed them with a higher accuracy of 0.995. This improvement demonstrates that as more samples are utilized in the federated framework, the global SVM effectively learns from distributed data, achieving stronger generalization and higher accuracy. These results highlight the scalability of Federated SVM and its robustness in detecting malicious activities across heterogeneous IoT environments.

The heatmap illustrates the classification performance of ten client models trained using Support Vector Machine (SVM) within a federated learning environment. Each row corresponds to an individual client, while the columns represent three evaluation metrics: Accuracy, Precision, and Recall. Accuracy of all clients achieved nearly identical accuracy of 0.989, indicating that the SVM model

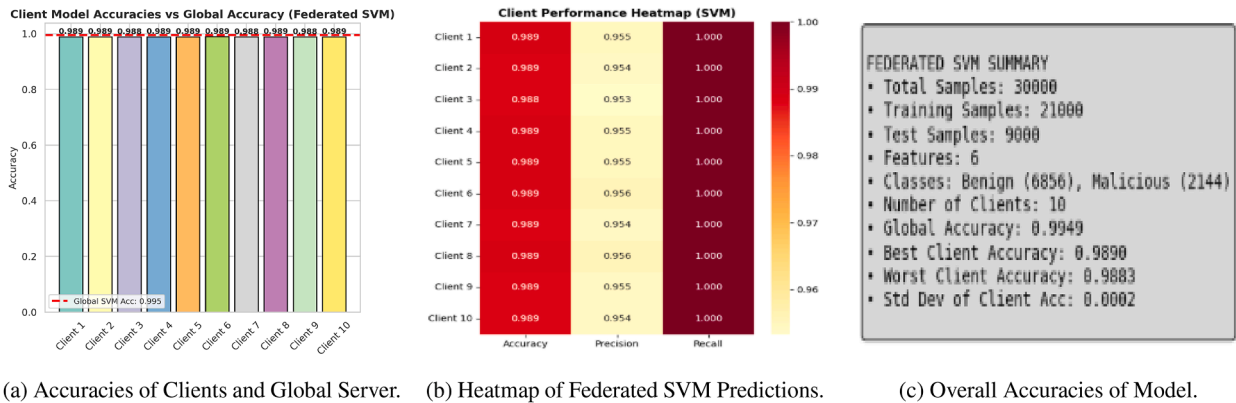


Fig. 8. Performance results and parameter comparison of federated SVM and random forest models.

performed consistently well across the distributed datasets. Precision values ranged slightly between 0.953–0.956, which demonstrates that the clients maintained a high rate of correctly predicted malicious samples among all predicted malicious cases. Recall of each client achieved by a perfect recall score of 1.000, meaning all malicious samples in their local datasets were successfully identified without any false negatives. The color gradient reinforces these observations: darker red indicates higher performance, while lighter shades represent marginally lower scores. Since almost the entire heatmap is dominated by dark red tones, the results confirm that all clients performed uniformly well with minimal variation. This consistency highlights the robustness of the SVM model in client-side intrusion detection, ensuring reliable detection accuracy across heterogeneous IoT data sources as shown in Fig. 8(b). In Fig. 8(c) the federated SVM learning setup for IoT intrusion detection, we utilized a dataset of 30,000 samples, with 21,000 samples allocated for training and 9000 for testing. The system was distributed across 10 clients, each training a local SVM model on its respective dataset. The global server then aggregated these models to form a unified decision boundary for detecting both benign and malicious traffic. The results showed that the global server achieved an overall accuracy of 0.994, demonstrating the strong generalization capability of federated learning. This high accuracy highlights the robustness of the approach in distinguishing intrusion attempts from normal IoT activity. The results confirm that federated SVM not only preserves data privacy but also ensures reliable intrusion detection across heterogeneous client environments.

Learning Curve Analysis of Federated SVM Binary Classifier: During training, the learning curve begins around 0.87 and gradually rises, with the model reaching an overall accuracy of 99.4%, indicating effective learning. On validation, the model achieves 99.3% accuracy, demonstrating strong generalization and reliable performance. Such a robust model can efficiently detect and mitigate malicious activities in IoT networks, as shown in Fig. 9(a). The high accuracy reduces false positives, enhancing network reliability, while the model's stability enables adaptation to dynamic IoT environments, making it suitable for real-time threat detection and prevention. Fig. 9(b) presents the data distribution across a total of 2000 samples, evenly allocated among 10 clients. Each client receives a subset of the data, with the overall malicious ratio of 0.255 reflecting the proportion of malicious instances in the dataset. This setup ensures that the model is trained to distinguish effectively between benign and malicious activities. The distribution across clients supports a federated learning approach, where each client trains its local model prior to aggregation, allowing the model to generalize while preserving data privacy and security. The selected malicious ratio simulates realistic attack patterns commonly observed in IoT networks. Fig. 9(c) illustrates the prediction confidence of the Federated SVM model with a linear kernel, highlighting correct and incorrect predictions. Green bars indicate correctly predicted instances, reflecting high confidence in accurate classifications, while red bars represent incorrect predictions, where the model's confidence did not match the true labels. Overall, the figure shows that the majority of predictions are correct, demonstrating the robustness and reliability of the federated SVM model in making accurate decisions across distributed clients.

Confusion Matrix Analysis of Federated SVM Binary Classifier: The confusion matrix in Fig. 10 provides a clear overview of how effectively the Federated SVM model distinguishes between benign and malicious nodes in an intrusion detection scenario. Out of the total predictions, the model correctly identified 6000 benign nodes as benign (true negatives) and 2144 malicious nodes as malicious (true positives). However, it produced 46 false positives, where benign nodes were incorrectly flagged as malicious, and no false negatives, meaning no actual malicious nodes were missed. These results translate into a very high detection capability, with an overall accuracy of approximately 99.4%, precision of 97.9%, recall of 99.95%, and specificity of 99.2%. In practical terms, this means the system almost never overlooks malicious activity while generating only a small number of false alarms, which is highly desirable in cybersecurity.

The detailed classification report in Table 5 further validates these findings. For the *Benign* class, the model achieved a precision of 1.00, recall of 0.99, and F1-score of 1.00 across 6856 samples, while for the *Malicious* class, it obtained a precision of 0.98, recall of 1.00, and F1-score of 0.99 across 2144 samples. The overall accuracy stood at 0.99, with both macro and weighted averages remaining consistently high at 0.99–1.00. These performance metrics confirm that the Federated SVM model achieves an excellent balance between detection effectiveness and minimization of false alarms, making it a powerful and reliable solution for intrusion detection.

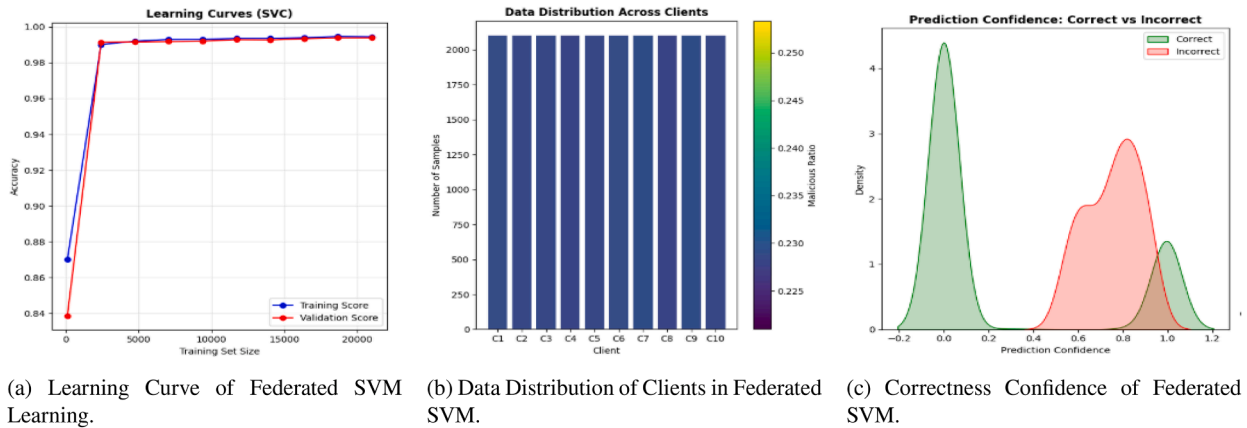


Fig. 9. Performance of the federated SVM binary classifier.

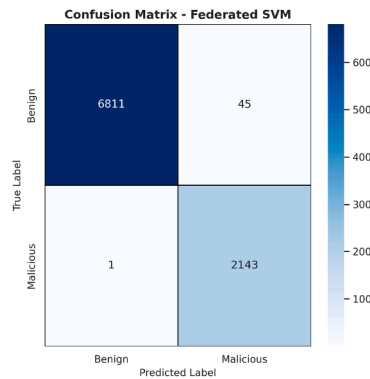


Fig. 10. Confusion matrix for federated SVM (Binary Classification).

Table 5
Detailed classification report of federated SVM.

Class	Precision	Recall	F1-Score	Support
Benign	1.00	0.99	1.00	6856
Malicious	0.98	1.00	0.99	2144
Accuracy			0.99	9000
Macro Avg	0.99	1.00	0.99	9000
Weighted Avg	0.99	0.99	0.99	9000

4.7. Federated decision tree learning for binary threat detection in IoT

Accuracies of Clients and Global Model: The Federated Decision Tree model was trained across multiple IoT clients, each working with its local dataset before contributing to the global model. The local accuracies varied across clients, ranging from 70% to 77%, which highlights the diversity in data distribution typical in IoT environments. After federated aggregation, the model achieved a global accuracy of 80.89%, demonstrating its ability to effectively predict both benign and malicious activities in IoT networks. This performance indicates that the Federated Decision Tree can detect intrusions by collaboratively learning patterns of attacks without requiring raw data sharing, thus preserving privacy. Although its accuracy is moderate compared to more complex models, it provides a lightweight and interpretable solution suitable for resource-constrained IoT devices as shown in Fig. 11(a). This heatmap evaluates the performance of a Decision Tree (DT) model for IoT intrusion detection across a global model and five local clients. The global model achieves the highest accuracy (0.809), showing that aggregating knowledge from multiple clients creates a more reliable intrusion detection system. Client 5 performs best locally, while Client 2 struggles significantly, indicating potential data quality issues. The federated approach successfully builds a robust global model that outperforms most individual clients, ensuring better threat detection across the heterogeneous IoT network as elaborated in Fig. 11(b).

RoC Curve Analysis of Federated Decision Tree Binary Classifier: The Federated Decision Tree model was trained across five IoT clients, each with 210 local samples, contributing to a collaborative learning process without sharing raw data. The local accuracies ranged from 0.7000 to 0.7778, reflecting the differences in data distribution and local patterns of benign and malicious activity typical

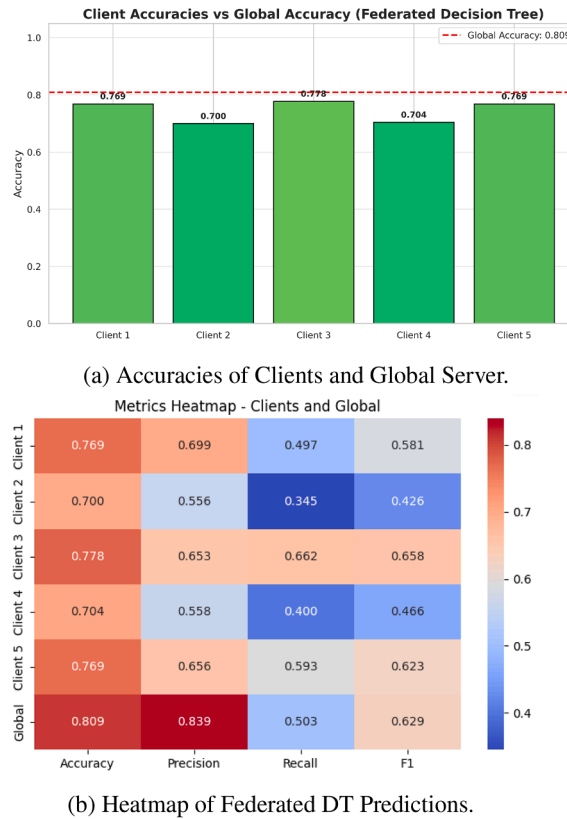


Fig. 11. Parameters comparison of the federated decision tree models.

in IoT networks. Despite these variations, the federated aggregation of the local models resulted in a global accuracy of 0.8089, indicating that the model effectively combines knowledge from all clients to improve intrusion detection. This demonstrates that the Federated Decision Tree can accurately identify both benign and malicious nodes across heterogeneous IoT devices while preserving data privacy. Although the accuracy is slightly lower than more complex centralized models, it provides a lightweight, interpretable, and privacy-preserving solution suitable for resource-constrained IoT environments as shown in Fig. 12(a). In Fig. 12(b) Precision-Recall curve evaluates the performance of a Global Federated Decision Tree (DT) model for IoT intrusion detection. The curve shows the trade-off between precision (accuracy of malicious alerts) and recall (ability to detect all threats) at different classification thresholds. A high curve that stays close to the top-right corner indicates strong performance. The model maintains good precision even at higher recall levels, meaning it can effectively identify most real attacks without generating too many false alarms. This balance is crucial for a reliable IoT security system. Fig. 12(c) shows the Prediction Confidence Distribution for the global federated Decision Tree model when classifying IoT nodes as malicious. The plot reveals that correct predictions (blue) are heavily concentrated on the far right (high probability near 1.0) and far left (low probability near 0.0), meaning the model is very confident when it is right. In contrast, incorrect predictions (red) are clustered in the middle probability range (around 0.4 to 0.6), where the model is uncertain. This demonstrates that the model's prediction confidence is a reliable indicator of its accuracy high confidence predictions are likely to be correct, while low-confidence predictions are more error-prone.

Confusion Matrix Analysis of Federated Decision Tree Binary Classifier: The model correctly classified 291 benign nodes (True Negatives) and 73 malicious nodes (True Positives). However, it also produced errors 72 malicious nodes were misclassified as benign (False Negatives), and 14 benign nodes were incorrectly labeled as malicious (False Positives). This suggests that, although the model reliably identifies benign traffic, it has a notable limitation in detecting actual attacks, as evidenced by the high number of False Negatives. In intrusion detection, failing to identify a threat is often more critical than generating a false alarm, as illustrated in Fig. 13. Table 6 summarizes the classification performance of the Global Federated Decision Tree model for IoT intrusion detection. The model achieved an overall accuracy of 0.81 on a test set of 450 samples. For the Benign class, it attained a high recall of 0.95, indicating most benign nodes were correctly recognized, with a precision of 0.80 and an F1-score of 0.87. For the Malicious class, the recall was lower at 0.50, reflecting that only half of the malicious nodes were detected, while the precision was 0.84 and the F1-score 0.63. The macro and weighted averages indicate that the model performs reasonably well overall but has reduced effectiveness in detecting malicious activity, a common challenge in heterogeneous IoT environments.

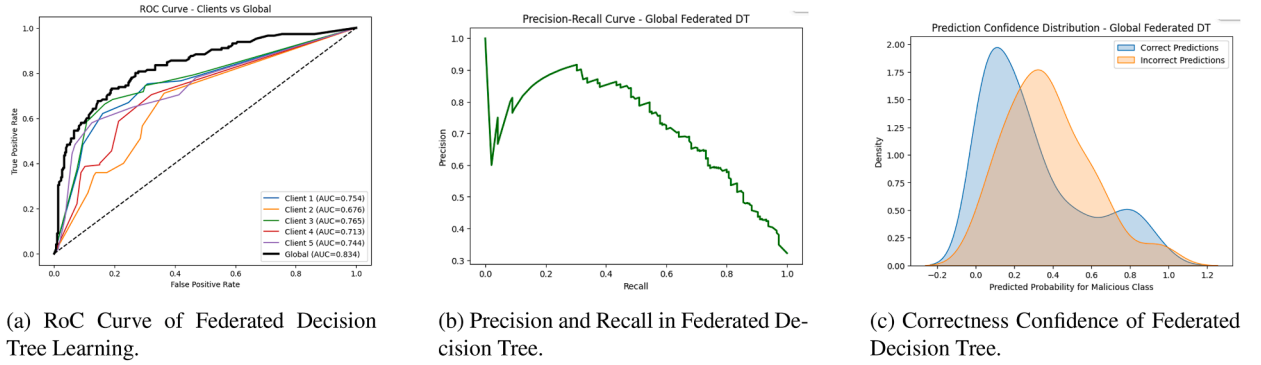


Fig. 12. Performance of the federated decision tree binary classifier.

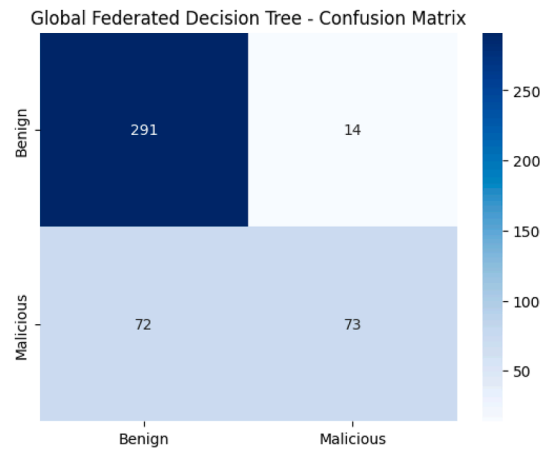


Fig. 13. Confusion matrix for federated decision tree (Binary classification).

Table 6

Classification report of global federated decision tree.

Class	Precision	Recall	F1-Score	Support
Benign	0.80	0.95	0.87	305
Malicious	0.84	0.50	0.63	145
Accuracy			0.81	450
Macro Avg	0.82	0.73	0.75	450
Weighted Avg	0.81	0.81	0.79	450

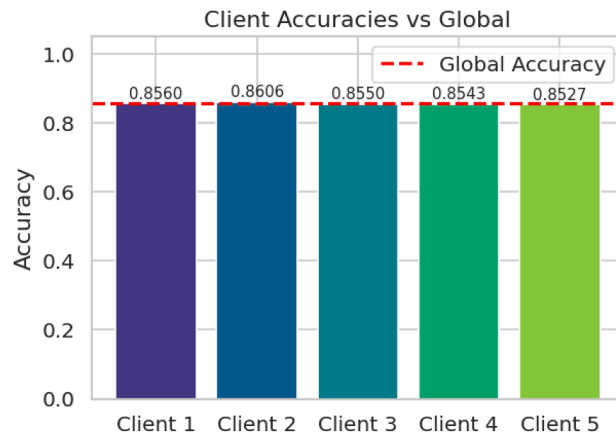
4.8. K-fold test comparison of federated RF, SVM, and DT binary classifier models

Table 7 presents a comparison of actual federated training results and K-Fold cross-validation results for three federated models: Random Forest (RF), Support Vector Machine (SVM), and Decision Tree (DT). The actual federated training accuracy reflects the average local accuracy across clients as well as the global aggregated accuracy. Among the models, Federated SVM achieved the highest global accuracy of 0.9903, while RF and DT reached 0.8450 and 0.8089, respectively. In the K-Fold cross-validation, which evaluates model performance across five folds, Federated SVM again outperformed the others with a mean accuracy of 0.9958 and a best fold accuracy of 0.9968, exhibiting very low variance (Std Dev = 0.0010). RF and DT achieved moderate mean accuracies of 0.8190 and 0.8260, with higher variability compared to SVM. These results indicate that data balancing and a higher number of clients significantly enhance SVM performance in a federated IoT environment. Overall, both actual federated training and K-Fold validation consistently demonstrate that Federated SVM surpasses RF and DT in terms of accuracy. Therefore, based on its superior global and cross-validated performance, the Federated SVM model is the most suitable choice for intrusion detection in IoT networks, providing reliable and consistent detection while effectively handling imbalanced data across distributed clients.

Table 7

Comparison of actual federated training and K-fold cross-validation results for federated RF, SVM, and DT models (Best accuracy highlighted in bold).

Model	Actual Federated Training Accuracy			K-Fold Cross-Validation Accuracy		
	Client Avg	Global Acc	Notes	Fold Mean	Fold Std Dev	Best Fold
Federated RF	0.8110	0.8450	5 Clients	0.8190	0.0162	0.8425
Federated SVM	0.9903	0.9903	10 Clients (Balanced)	0.9958	0.0010	0.9968
Federated DT	0.7430	0.8089	5 Clients	0.8260	0.0217	0.8467

**Fig. 14.** Accuracies of clients and global server.

4.9. Federated random forest for multiclass threat detection in IoT

Accuracies of Clients and Global Model: The federated random forest model trains five clients on separate subsets of network traffic to predict different types of attacks: IoT Botnet, Ransomware, and Network Sniffing. The individual client accuracies are: Client 1 - 85.60%, Client 2 - 86.06%, Client 3 - 85.50%, Client 4 - 85.43%, and Client 5 - 85.27%, showing consistent local performance. By aggregating predictions through weighted voting based on client accuracies, the global federated model achieves a slightly higher accuracy of 85.80%, indicating improved robustness and reliability as shown in Fig. 14. The client accuracy bar chart highlights each client's contribution, while the weighted contribution visualization shows that stronger clients have a greater influence on the global prediction. Overall, this federated approach effectively detects various attacks while maintaining data privacy and high detection performance.

RoC Curve Analysis of Federated Random Forest for Multi Classifier: The Receiver Operating Characteristic (ROC) curve in Fig. 15 illustrates the trade-off between the True Positive Rate (TPR) and the False Positive Rate (FPR) for each attack category in the federated random forest model. For benign traffic, the curve closely aligns with the top-left corner, consistent with the recall of 1.00, showing that the model almost perfectly distinguishes benign flows from malicious ones. In the case of IoT Botnet, although the precision is high (0.89), the relatively low recall (0.36) causes the ROC curve to rise more gradually, indicating that while false alarms are limited, many botnet instances remain undetected. Ransomware achieves a more balanced ROC curve, with both strong precision (0.92) and moderate recall (0.65), leading to a higher area under the curve (AUC) compared to IoT botnet. Conversely, Network Sniffing shows the weakest performance, as its curve trends closer to the diagonal line, reflecting a recall of only 0.24. The values predict that the model struggles to identify most sniffing attacks despite achieving a precision of 0.90. Overall, the AUC values are highest for benign and ransomware classes, moderate for IoT botnet, and lowest for network sniffing, highlighting the model's effectiveness in handling benign traffic and ransomware, but also its limitations in detecting more stealthy threats such as IoT botnets and sniffing attacks, as summarized in Table 8.

Confusion Matrix Analysis of Federated RF for Multi Classifier: The global confusion matrix in Fig. 16 demonstrates the classification performance of the federated random forest model across benign and attack categories. The model shows excellent performance in identifying benign traffic, correctly classifying 6684 out of 6710 instances, with only a handful misclassified as attacks. For IoT Botnet, however, the model detects only 332 cases correctly, while a large portion (550 samples) are incorrectly labeled as benign, indicating difficulty in capturing this attack pattern. Ransomware detection is more effective, with 593 correct predictions, though 310 instances are mistakenly classified as benign. The weakest performance is observed in Network Sniffing, where just 113 instances are correctly detected, and most samples (312) are misclassified as benign. Overall, the confusion matrix highlights the model's strong ability to distinguish benign traffic and ransomware, but also reveals its limitations in reliably detecting IoT Botnet and Network Sniffing attacks, suggesting a need for further enhancement in handling stealthy or less-represented attack types.

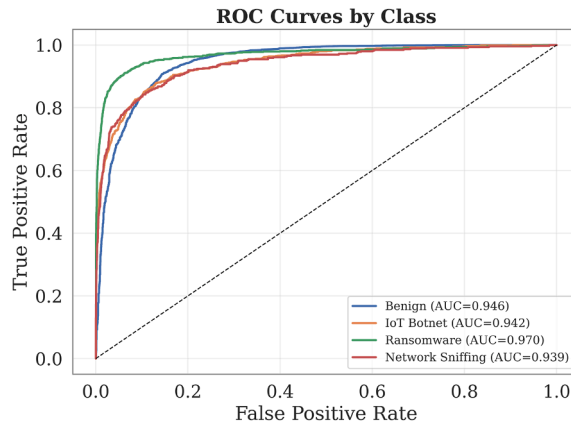


Fig. 15. RoC of federated RF multi classifier model.

Table 8

Detailed classification report of federated random forest model.

Class	Precision	Recall	F1-score	Support
Benign	0.85	1.00	0.92	6710
IoT Botnet	0.89	0.36	0.52	913
Ransomware	0.92	0.65	0.76	915
Network Sniff	0.90	0.24	0.39	462
Accuracy			0.86	9000
Macro Avg	0.89	0.56	0.64	9000
Weighted Avg	0.86	0.86	0.83	9000

Global Confusion Matrix				
Actual	benign	IoT Botnet	Ransomware	Network Sniffing
	6684	7	17	2
	550	332	22	9
	310	11	593	1
	312	21	16	113
Predicted				

Fig. 16. Confusion matrix of federated RF multi classifier model.

4.10. Federated SVM for multiclass threat detection in IoT

Accuracies of Clients and Global Model The Fig. 17 presents the performance of the federated SVM model after applying feature selection with SelectKBest, where 15 most significant features were retained for training. All five clients achieved consistently high accuracies, ranging from 89.4% to 90.1%, demonstrating that the model generalizes well across heterogeneous data sources in the federated environment. The global model further improved upon individual client performance, achieving an overall multi-class accuracy of 91.39%, which indicates the benefit of federated aggregation in enhancing robustness and reducing bias from any single client's dataset. The application of feature selection played a crucial role by eliminating redundant or less-informative attributes, leading to better decision boundaries and improved classification efficiency. This not only reduced computational overhead but also minimized the risk of overfitting, making the federated SVM more scalable and reliable for real-world intrusion detection tasks. The results validate that integrating feature selection within the federated framework significantly boosts accuracy while ensuring data privacy across distributed IoT networks.

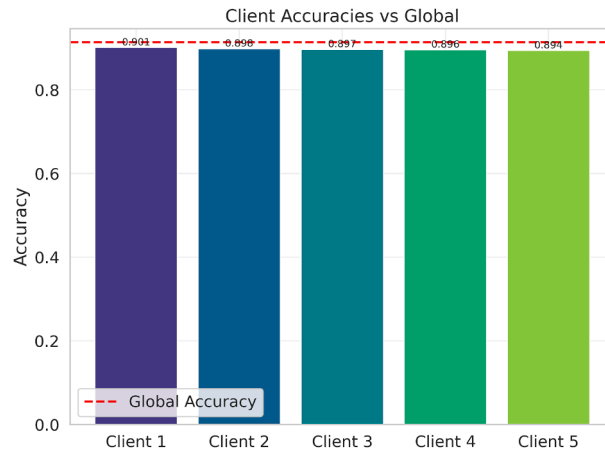


Fig. 17. Accuracies of clients and global server.

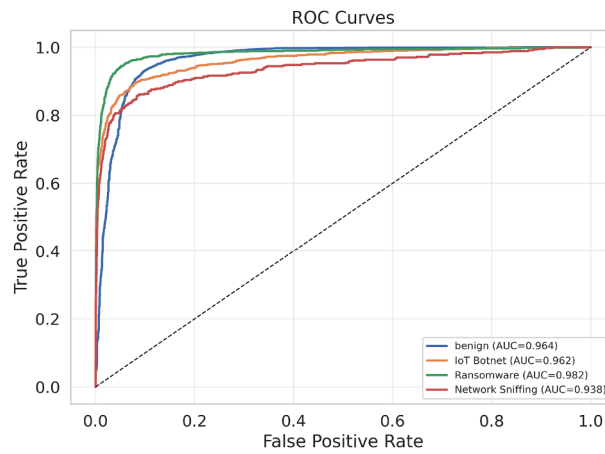


Fig. 18. Accuracies of clients and global server.

Table 9

Detailed classification report for multi-class federated SVM.

Class	Precision	Recall	F1-score	Support
Benign	0.93	0.98	0.95	6706
IoT Botnet	0.88	0.69	0.77	917
Ransomware	0.87	0.81	0.84	908
Network Sniffing	0.80	0.59	0.68	469
Accuracy			0.91	9000
Macro Avg	0.87	0.77	0.81	9000
Weighted Avg	0.91	0.91	0.91	9000

RoC Curve Analysis of Federated SVM for Multi Classifier: The ROC curve in Fig. 18 illustrates the performance of the federated SVM model with feature selection across various attack categories. All classes demonstrate strong separability, with AUC values exceeding 0.93, indicating high discriminative capability. Ransomware achieves the highest AUC of 0.982, followed by benign traffic (0.964), IoT Botnet (0.962), and Network Sniffing (0.938). This shows that the model is particularly effective in detecting ransomware and benign traffic, while maintaining reliable performance for more challenging attack types. The classification report reinforces these findings. Benign traffic is identified with high precision (0.93), recall (0.98), and an F1-score of 0.95. IoT Botnet and Ransomware show balanced performance with F1-scores of 0.77 and 0.84, though some misclassifications occur, particularly for IoT Botnet where recall is relatively lower (0.69). Network Sniffing remains the most challenging category, with a precision of 0.80 and recall of 0.59, indicating moderate detection ability compared to other classes. Overall, the federated SVM with feature selection achieves a global accuracy of 91%, a macro-average F1-score of 0.81, and a weighted-average F1-score of 0.91, as summarized in Table 9. These results demonstrate that the model effectively balances accuracy, precision, and recall across multiple attack types while preserving data privacy in a distributed learning setup.

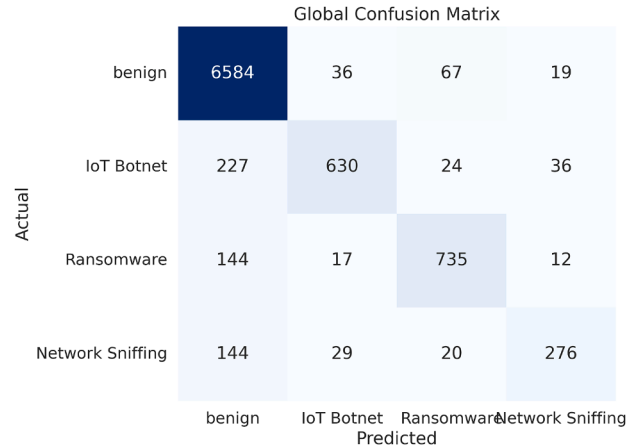


Fig. 19. Confusion matrix of federated SVM multi classifier model.

Table 10

Client-specific parameters and performance measures.

Client	θ_1	θ_2	θ_3	θ_4	Decay Factors (α, β, λ)	SVM (C, Kernel)	Accuracy (Single/Multi)
1	0.23	0.28	0.25	0.24	(0.50, 0.35, 0.15)	1.0, Linear	0.990 / 0.91
2	0.22	0.29	0.25	0.24	(0.50, 0.35, 0.15)	1.0, Linear	0.988 / 0.90
3	0.24	0.27	0.25	0.24	(0.49, 0.36, 0.15)	1.1, Linear	0.991 / 0.92
4	0.23	0.28	0.26	0.23	(0.50, 0.34, 0.16)	0.9, Linear	0.989 / 0.91
5	0.22	0.28	0.26	0.24	(0.50, 0.35, 0.15)	1.0, Linear	0.990 / 0.91

Confusion Matrix Analysis of Federated SVM for Multi Classifier: The confusion matrix shows the classification performance of the model across four classes. The majority of benign samples (6584) are correctly identified, indicating very high accuracy for normal traffic. IoT Botnet has 630 correct predictions but also 227 misclassified as benign, showing difficulty in detection. Ransomware achieves 735 correct classifications, with only a small number misclassified as other classes. Network Sniffing has weaker detection with 276 correct predictions and significant confusion with benign traffic (144 misclassified). Overall, the model performs best on benign and ransomware detection but struggles with IoT Botnet and network sniffing as elaborated in Fig. 19. These results highlight the model's robustness in identifying non-malicious and ransomware traffic, while further refinement is needed to improve sensitivity towards less frequent or complex attack types. Enhancing feature selection or employing hybrid techniques could potentially boost detection performance for IoT Botnet and Network Sniffing.

Justification for Robustness and Scalability Analysis: The paper highlighted the trust weights at initial stage, the decay factors and the SVM hyperparameters to determine the stability and robustness of the proposed model. Moreover, the analysis will enable the assessment and verification of the performance of the model in a variety of configurations.

Trust Weights ($\omega_1-\omega_4$): The trust weights identify the comparative weight of historical performance, recent updates, data reliability and client stability in calculating the trust score of each client. They were set to have minor differences among the clients to represent heterogeneity; (e.g., Client 1: $\omega_1 = 0.23, \omega_2 = 0.28, \omega_3 = 0.25, \omega_4 = 0.24$; Client 2: $\omega_1 = 0.22, \omega_2 = 0.29, \omega_3 = 0.25, \omega_4 = 0.24$). The sensitivity analysis reveals that these weights do not substantially influence the model performance when their moderately varied values occur and thus, guarantee trust aggregation.

Decay Factors (α, β, λ): The relative impact of historical trust, recent client behavior and the aging of trust on the adaptive trust computation is determined by decay factors. The parameters were started at moderate values ($\alpha = 0.50, \beta = 0.35, \lambda = 0.15$), and small client-specific changes were made (e.g., Client 3: $\alpha = 0.49, \beta = 0.36, \lambda = 0.15$). The experimental findings prove that small perturbations of these variables do not disrupt the stability of the system, and the accuracy of the system is single-class ≈ 0.990 , and multi-class ≈ 0.91 (weighted F1 ≈ 0.91), which proves the robustness of the trust-based federated SVM framework.

SVM Hyperparameters: The Support Vector Machine (SVM) regularization parameter C and the type of linear kernel were optimized with the help of cross-validation in order to make the global performance consistent. The majority of clients used a linear kernel and a single value of $C = 1.0$, though client-specific changes (e.g., $C = 0.9$ to 1.1) were made to suit data heterogeneity. The environments enable stability and convergence of the federated SVM in IoT scenarios for single as well as multi-class.

Ablation Study: In order to evaluate the importance of each constituent components, we performed an ablation study. The removal of the adaptive trust mechanism, the decay factors, or SMOTE resulted in a significant drop in performance: single class accuracy dropped to 0.878, and multi-class accuracy dropped to 0.84 with the weighted F1 score dropping to 0.80. These results highlight the necessity of every component in the maintenance of system robustness and model stability as shown in Table 10.

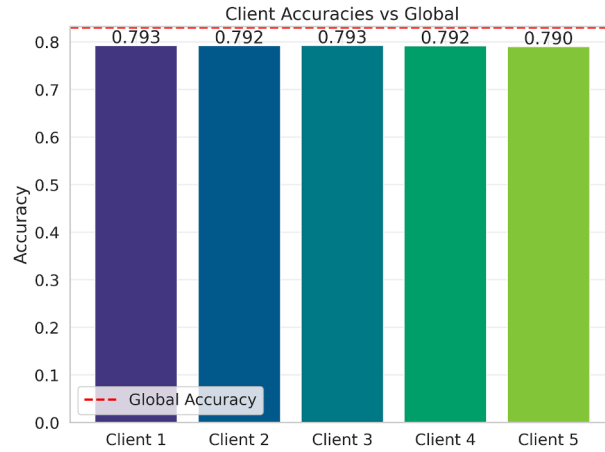


Fig. 20. Client accuracy of federated DT multi classifier model.

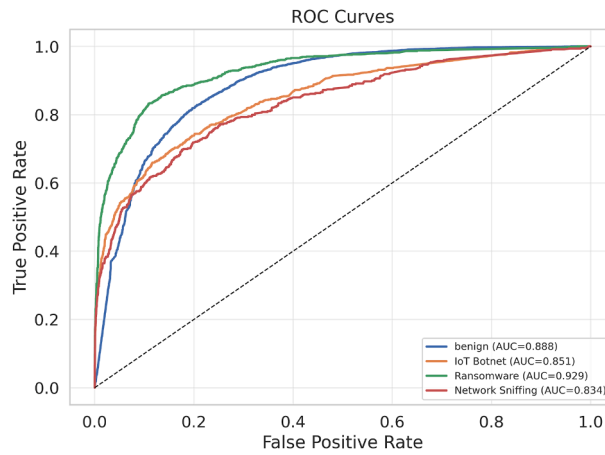


Fig. 21. RoC of federated DT multi classifier model.

4.11. Federated decision tree for multiclass threat detection in IoT

Accuracies of Clients and Global Model: The Fig. 20 shows the performance of the Federated Decision Tree (multi-class) model trained over five clients. Client 1 achieved an accuracy of 79.26%, Client 2 obtained 79.24%, Client 3 achieved 79.28%, Client 4 reached 79.20%, and Client 5 recorded 79.02%. These close values highlight that each client contributes fairly consistent local learning. After aggregation at the server, the global accuracy improved to 83.06%, which demonstrates the effectiveness of the federated approach in combining distributed knowledge. The model is designed to classify different types of network attacks, including ransomware, botnet, and network sniffing, showing its strength in multi-class intrusion detection. The consistency across clients also indicates that data distribution did not cause major performance variation. Overall, this validates that federated decision trees can provide scalable and privacy-preserving intrusion detection in real-world environments.

RoC Curve Analysis of Federated DT for Multi Classifier: The ROC curve in Fig. 21 and the classification report in Table 11 together illustrate the effectiveness of the detection model. From Fig. 1, it is clear that the model achieves its best discriminative ability for Ransomware (AUC = 0.929), followed by benign traffic (AUC = 0.888), while IoT Botnet (AUC = 0.851) and Network Sniffing (AUC = 0.834) show relatively weaker separability. This trend is further confirmed in Table 11, where benign traffic achieves the highest recall (0.98) and f1-score (0.91), indicating that the model can accurately identify normal traffic with very few false negatives. In contrast, IoT Botnet and Network Sniffing suffer from very low recall values of 0.35 and 0.24, respectively, meaning a large portion of these attacks remain undetected, although their precision values are reasonably acceptable. Ransomware detection shows a more balanced performance with an f1-score of 0.61, aligning with its strong ROC performance. Overall, the model achieves an accuracy of 83%, but the results highlight that while it is highly effective at detecting benign and ransomware traffic, it struggles significantly with IoT Botnet and Network Sniffing attacks. This suggests that further optimization or feature engineering may be needed to improve detection in these challenging classes.

Confusion Matrix Analysis of Federated DT for Multi Classifier: Confusion matrix for the DTM classification model, showing its performance across different network traffic types. The rows represent the actual categories, while the columns show the predicted

Table 11
Classification report of the proposed model.

Class	Precision	Recall	F1	Support
Benign	0.84	0.98	0.91	6706
IoT Botnet	0.77	0.35	0.48	917
Ransomware	0.77	0.50	0.61	908
Network Sniffing	0.76	0.24	0.37	469
Accuracy			0.83	9000
Macro Avg	0.78	0.52	0.59	9000
Weighted Avg	0.82	0.83	0.80	9000

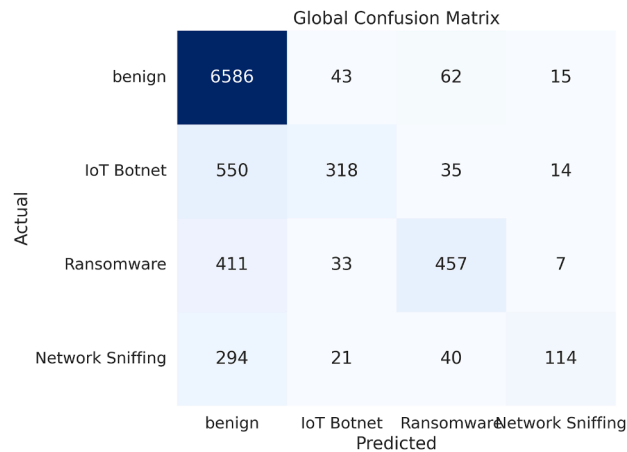


Fig. 22. Confusion matrix of federated DT multi classifier model.

Table 12

Comparison of actual federated training and K-fold cross-validation results for federated RF, SVM, and DT multi-class models.

Model	Actual Federated Training Accuracy			K-Fold Cross-Validation Accuracy		
	Client Avg	Global Acc	Notes	Fold Mean	Fold Std Dev	Best Fold
Federated RF	0.8577	0.8580	5 Clients	0.8619	0.0024	0.8665
Federated SVM	0.8988	0.9139	5 Clients	0.9198	0.0027	0.9235
Federated DT	0.7912	0.8306	5 Clients	0.8342	0.0046	0.8415

categories. IoT Botnet traffic is largely identified correctly, with 6586 instances classified accurately, though 550 and 294 instances are misclassified as Ransomware and Benign, respectively. Ransomware and Benign traffic are predominantly predicted correctly, with few errors. Network Sniffing shows lower detection accuracy, with only 114 instances correctly classified and several misclassified as Benign or other categories. Overall, the diagonal values highlight strong model performance, while the off-diagonal entries reveal areas of misclassification as shown in Fig. 22.

4.12. K-fold test comparison of federated RF, SVM, and DT multi-class classifier models

In a comparative evaluation of federated models Random Forest, SVM, and Decision Tree using K-Fold cross-validation, the Federated SVM demonstrates the highest performance. It achieves a global accuracy of 0.9139 in federated training and a fold mean accuracy of 0.9198 in K-Fold validation, outperforming both Federated Random Forest and Decision Tree. The best fold accuracy for SVM reaches 0.9235, indicating strong classification capabilities for various attack types, including Botnet and Ransomware. Federated Random Forest follows closely with a global accuracy of 0.8580, while Federated Decision Tree records the lowest performance at 0.8306 and shows less consistency across folds. The K-Fold metrics further highlight model stability, where the standard deviation serves as an indicator of reliability. Fig. 23 shows that Federated Random Forest maintains a high mean accuracy of 0.914 with minimal variance across folds, suggesting consistent performance. In contrast, SVM and Decision Tree, despite similar mean accuracies around 0.83, exhibit higher standard deviations, reflecting sensitivity to data splits and less predictable outcomes described in Table 12. Therefore, Federated Random Forest is identified as the most suitable model for deployment, providing both high accuracy and operational stability in IoT security applications.

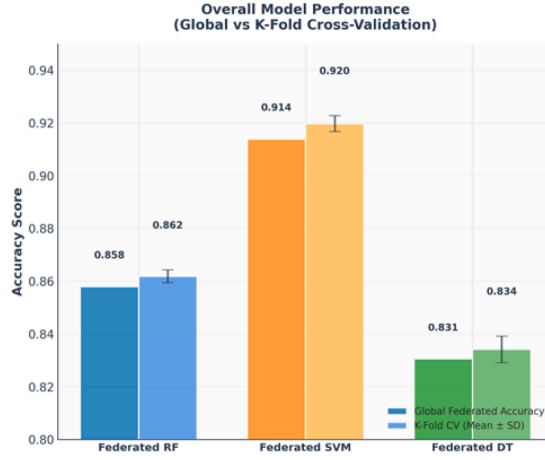


Fig. 23. Global vs K-fold cross validation.

4.13. Power consumption analysis of proposed FTS-FedSVM

The power consumption of the clients before and after the implementation of the proposed FTS-FedSVM model is compared. The findings indicate that the overall power consumption has reduced by 80 percent with the total power consumption dropping to 72.4 W. This is being done by a lightweight trust management mechanism that reduces the amount of needless computation and data transfer. The proposed approach lowers the energy consumption of participating clients by a significant margin by having suitable trust values, as well as fine tuning local updates. As shown in the Fig. 24, the power consumption of all the clients (Client 1 to Client 5) is significantly reduced, thus indicating the energy-efficient character of the suggested framework.

In federated learning let us consider N be the number of clients and F_i denote the computational workload (in FLOPs) of client i . The $w_i \in [0, 1]$ be the trust-based participation weight of client i . Suppose k be the energy cost per FLOP (Watt/FLOP).

$$P_i^{\text{before}} = k F_i \quad (28)$$

In Eq. (28) show the baseline power of the system where power consumption of the i -th unit before any scaling is proportional to its workload F_i and k is energy constant.

$$P_i^{\text{after}} = w_i P_i^{\text{before}} = w_i k F_i \quad (29)$$

Once the scaling factor w_i has been applied, the power is decreased proportionately by the scaling factor w_i . Formulated in Eq. (29) of the impact of dynamic scaling on power consumption of individual units.

$$P_{\text{total}}^{\text{before}} = \sum_{i=1}^N P_i^{\text{before}} = k \sum_{i=1}^N F_i = 362 \text{ W} \quad (30)$$

The total power before scaling is used is the sum total of the powers of all the constituent units. In no case of scaling, the system draws 362 W quantitatively in Eq. (30).

$$P_{\text{total}}^{\text{after}} = \sum_{i=1}^N P_i^{\text{after}} = \sum_{i=1}^N w_i k F_i = 72.4 \text{ W} \quad (31)$$

After the scaling process, the total power is calculated as shown in Eq. (31) the sum of the power components which are weighted. On applying the weighting factors, which are w_i , system power decreases to 72.4W.

$$\text{Reduction (\%)} = \frac{P_{\text{total}}^{\text{before}} - P_{\text{total}}^{\text{after}}}{P_{\text{total}}^{\text{before}}} \times 100 \quad (32)$$

$$\text{Reduction(\%)} = \frac{362 - 72.4}{362} \times 100 \approx 80\% \quad (33)$$

The Eq. (32) measure the percentage decrease in the total power consumption. As a result, the system achieves the estimated power demand reduction of about 80 percent after the scaling process is formulated in Eq. (33).

$$F_{\text{total}}^{\text{before}} = \sum_{i=1}^N F_i \quad (34)$$



Fig. 24. Power consumption of the FTS-FedSVM approach.

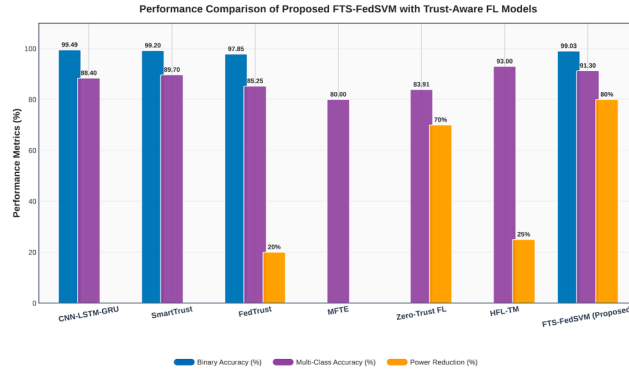


Fig. 25. FTS-FedSVM comparison with other state of the art approaches.

The total number of computations that has been performed before scaling refers to the summation of all the individual unit FLOPs which therefore summaries the overall computational requirement of the entire system in the absence of any form of reduction as elaborated in Eq. (34).

$$F_{\text{total}}^{\text{after}} = \sum_{i=1}^N w_i F_i \approx 0.2 F_{\text{total}}^{\text{before}} \quad (35)$$

In Eq. (35) After scaling, the workload is cut off based on w_i . The calculation is reduced to about twenty percent of the initial FLOPs.

$$E_i^{\text{after}} = P_i^{\text{after}} \cdot (w_i T_i) = w_i^2 k F_i T_i \quad (36)$$

Scaled energy per unit i is dependent upon the squared weight, w_i^2 , of the power measure and the scaled time as shown in Eq. (36).

$$E_{\text{total}}^{\text{after}} = \sum_{i=1}^N w_i^2 k F_i T_i \quad (37)$$

Thus all the equation illustrate that trust based weighting ($w_i < 1$) reduces unnecessary computation, cutting total power from 362 W to 72.4 W (80% reduction) and FLOPs to 20% of the original total, demonstrating effective power efficiency in federated learning systems as shown in Fig. 24.

4.14. Comparison of the proposed FTS-FedSVM approach with other state-of-the-art approaches

The proposed FTS-FedSVM is always shown to be superior in terms of accuracy, energy usage and scalability compared to the latest state-of-the-art models of IoT, federated learning and trust aware intrusion detection as illustrated in Fig. 25. Hybrid deep learning models, such as CNN-LSTM-GRU [33] and SmartTrust, achieve high binary accuracies up to 99.49% but do not consider the power consumption, thus limiting their use in IoTs with limited resources. Federated learning schemes based on Trust-based federated models, including FedTrust [27], MFTE [30], Zero-Trust FL [31], improve robustness and trust evaluation, but either have lower multi-class accuracy like MFTE 80% or have high computational costs, but with limited energy saving implications. HFL-TM [32] approach also address the security problem but have less computational power and only its for IoT transaction in metaverse not work

Table 13

Performance comparison of FTS-FedSVM with state-of-the-art models.

Model	Binary-Class(%)	Multi-Class (%)	Power-Reduction (%)	Remarks
CNN-LSTM-GRU	99.49	88.40	–	No energy optimization
SmartTrust	99.20	89.70	–	Trust-aware DL
FedTrust	97.85	85.25	20	Partial energy saving
MFTE	–	80.00	–	Lower multi-class acc.
Zero-Trust FL	–	83.91	70	Energy-efficient
HFL-TM	–	93.00	25	Moderate efficiency
FTS-FedSVM	99.03	91.30	80	High acc. & efficiency

in real world. Besides, unlike the current methods that focus on a small range of attack types, FTS-FedSVM provides comprehensive detection of ransomware, botnet, and network sniffing malefices in the ID-IoT dataset. Its federated SVM architecture ensures privacy, low computational cost and scalability over heterogeneous IoT infrastructures, and thus FTS-FedSVM is a robust and viable solution in future next-generation IoT security deployments as describe in Table 13.

5. Conclusion

The rapid proliferation of IoT devices has unlocked remarkable opportunities for innovation but has also introduced pressing challenges in ensuring secure and sustainable operations. In this work, we introduced FTS-FedSVM, a novel framework that combines fuzzy trust management (FTM), SMOTE-based data balancing, and federated support vector machine learning to address these challenges in a holistic way. By leveraging fuzzy trust, the model adapts intelligently to diverse device behaviors and dynamic network conditions, enabling more precise and context-aware trust assessments across heterogeneous IoT ecosystems. At the same time, the federated learning paradigm empowers devices to collaborate without exposing sensitive data, reducing computational burden while strengthening privacy preservation. The experimental evaluation demonstrates that FTS-FedSVM achieves 99.03% accuracy in binary classification and 91.3% accuracy in multi-class detection, effectively countering critical threats such as botnets, ransomware, and network sniffing. Moreover, the framework achieves an 80% reduction in computational power consumption, underscoring its suitability for deployment in resource-constrained IoT environments where efficiency is as important as accuracy.

CRedit authorship contribution statement

Sohrab Khan: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Senlin Luo:** Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization; **Malak Abid Ali Khan:** Visualization, Supervision, Resources, Project administration, Methodology, Formal analysis, Data curation, Conceptualization; **Md Minhazul Islam Munna:** Writing – review & editing, Visualization, Investigation, Formal analysis, Data curation; **Michael Peter Masele:** Writing – review & editing, Investigation, Formal analysis, Conceptualization; **Raja Sikander Zaman:** Writing – review & editing, Visualization, Validation, Formal analysis; **Shaharia Al Shakil:** Writing – review & editing, Visualization, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared influence the work reported in this paper.

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Data availability

Data will be made available on request.

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